ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^3 = 27$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q1/5

1601

QUESTION ID

S52-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\sqrt[3]{125}$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q2/5

1602

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $5^3 = 125$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is $\sqrt[3]{64}$?

A. 21.3

B. 4

C. 8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q3/5

1603

QUESTION ID

S52-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q3/5

True or false?

$\sqrt[3]{(-27)} = -3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q4/5

1604

QUESTION ID

S52-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-3)^3 = -27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q4/5

True or false?

$\sqrt{16} = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q5/5

1605

QUESTION ID

S52-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^2 = 16$, and $\sqrt{}$ means the nonnegative root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSF-BF.B.4.a

FORWARD 1

Verify inverses by composition • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Show that $f(x) = 2x^3$ and $g(x) = \sqrt[3]{(x \div 2)}$ are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q1/5

1607

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$f(g(x)) = 2 \cdot (x \div 2) = x$ and $g(f(x)) = \sqrt[3]{(x^3)} = x$.

GRADING NOTE Form required: both compositions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F1-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = (x + 1) ÷ (x − 1).

Find f(f(3)).

f(f(3)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q2/5

1608

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. f(3) = 2 and f(2) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q2/5

Which function is its own inverse?

A. $f(x) = 1 \div x$

B. $f(x) = 2x$

C. $f(x) = x + 1$

D. $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q3/5

1609

QUESTION ID

S52-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $f(x) = 1 \div x$

GRADING NOTE $1 \div (1 \div x) = x$ for $x \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q3/5

True or false?

If f(g(x)) = x on the domain of g and g(f(x)) = x on the domain of f, then g is the inverse of f.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q4/5

1610

QUESTION ID

S52-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the composition test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q4/5

True or false?

f(x) = x + 3 and g(x) = x − 3 are not inverses, because f(g(1)) = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q5/5

1611

QUESTION ID

S52-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(g(1)) = f(−2) = 1. They are inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSF-BF.B.4.a

FORWARD 2

Restrict a domain to produce an inverse • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

Restrict the domain of $f(x) = x^2 + 2$ so that f has an inverse, and give the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q1/5

1613

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 0$;

$f^{-1}(x) = \sqrt{(x - 2)}$

GRADING NOTE Form required: the restriction and the inverse. $x \leq 0$ with $f^{-1}(x) = -\sqrt{(x - 2)}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why f(x) = |x| has no inverse on all real numbers.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q2/5

1614

QUESTION ID
S52-F2-Q2

TYPE
Short answer

CORRECT ANSWER
f(3) = f(−3) = 3, so the input of 3 for an inverse would need two outputs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q2/5

$f(x) = (x + 4)^2$.

What is the largest domain that contains $x = 0$ on which f is one-to-one?

A. All real numbers

B. $-8 \le x \le 0$

C. $x \ge 0$

D. $x \ge -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q3/5

1615

QUESTION ID

S52-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x \ge -4$

GRADING NOTE f is increasing for $x \ge -4$. On $-8 \le x \le 0$, $f(-8) = f(0) = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^2$ restricted to $x \leq 0$ has the inverse $f^{-1}(x) = -\sqrt{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q4/5

1616

QUESTION ID
S52-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE For $x \leq 0$, $-\sqrt{(x^2)} = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q4/5

True or false?

$f(x) = x^2 + 1$ is one-to-one on all real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q5/5

1617

QUESTION ID

S52-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(1) = f(-1) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q5/5

Prove that linear functions grow by equal differences and exponential functions grow by equal factors over equal intervals

MAIN	Algebra I main question	**HSF-LE.A.1.a**
BACKWARD 1	Tell linear from nonlinear functions	**8.F.A.3**
BACKWARD 2	Compare average rates of change over intervals	**HSF-IF.B.6**
FORWARD 1	Derive the formula for the sum of a finite geometric series	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to turn equal factors into equal differences	**HSF-BF.B.5**

SET 53 • HSF-LE.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 3x + 5. Show that f grows by equal differences over equal intervals: find f(x + h) − f(x) in simplest form, and explain why it does not depend on x.

f(x + h) − f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q1/5

1620

QUESTION ID

S53-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

3h

GRADING NOTE REQUIRED WORK: Algebra showing f(x + h) − f(x) = 3(x + h) + 5 − (3x + 5) = 3h, and the statement that 3h depends only on the interval length h.

Form required: the simplified difference and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q1/5

$g(x) = 4 \cdot 2^x$. Show that g grows by equal factors over equal intervals: find $g(x + h) \div g(x)$ in simplest form, and explain why it does not depend on x.

$g(x + h) \div g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q2/5

1621

QUESTION ID

S53-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

2^h

GRADING NOTE REQUIRED WORK: Algebra showing $4 \cdot 2^{x+h} \div (4 \cdot 2^x) = 2^h$, and the statement that 2^h depends only on h.

Form required: the simplified ratio and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q2/5

A table shows x = 0, 2, 4, 6 and y = 5, 11, 17, 23.

Is the relationship linear or exponential? Explain using equal intervals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q3/5

1622

QUESTION ID

S53-M-Q3

TYPE

Short answer

CORRECT ANSWER

Linear: over each interval of 2, y increases by the same amount, 6.

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 80, 40, 20, 10.

Which description is correct?

A. Linear: y decreases by 40 each time

B. Exponential: y is multiplied by $\frac{1}{2}$ each time x increases by 1

C. Linear: y decreases by 20 each time

D. Neither linear nor exponential

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q4/5

1623

QUESTION ID

S53-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Exponential: y is multiplied by 1/2 each time x increases by 1

GRADING NOTE The differences are −40, −20, −10 (not equal), but the ratios are all $\frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q4/5

True or false?

A table with y-values 2, 5, 10, 17 for x = 0, 1, 2, 3 is linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q5/5

1624

QUESTION ID

S53-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The differences 3, 5, 7 are not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q5/5

SET 53 • HSF-LE.A.1.a

BACKWARD 1

Tell linear from nonlinear functions • Grade 8 • Functions • 8.F.A.3

Is $y = x^2 + 1$ linear? Use the points (0, 1), (1, 2), and (2, 5) to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q1/5

1626

GRADING NOTE Form required: the answer and the reason.

QUESTION ID

S53-B1-Q1

TYPE

Short answer

CORRECT ANSWER

No. y increases by 1 and then by 3, so the rate is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is y = 4 − 2x linear? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q2/5

1627

GRADING NOTE Form required: the answer and the reason.

QUESTION ID

S53-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. It has the form y = mx + b with slope −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q2/5

Which function is linear?

A. $y = 2^x$

B. $y = x^2$

C. $y = 0.5x - 3$

D. $y = 1 \div x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q3/5

1628

QUESTION ID

S53-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 0.5x - 3$

GRADING NOTE Only $y = 0.5x - 3$ has the form $y = mx + b$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q3/5

True or false?

The graph of a linear function is a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q4/5

1629

QUESTION ID

S53-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is why it is called linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q4/5

True or false?

The area of a square, $A = s^2$, is a linear function of s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q5/5

1630

QUESTION ID

S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its graph curves; the area grows by 1, 3, 5, ... for s = 0, 1, 2, 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSF-LE.A.1.a

BACKWARD 2

Compare average rates of change over intervals • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 3x + 5$.

Find the average rate of change from $x = 0$ to $x = 10$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q1/5

1632

QUESTION ID

S53-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 4 \cdot 2^x$.

Find the average rate of change from $x = 0$ to $x = 1$, and from $x = 1$ to $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q2/5

1633

QUESTION ID

S53-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4 and 8

GRADING NOTE Form required: both rates. (8 − 4) ÷ 1 and (16 − 8) ÷ 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q2/5

What is the average rate of change of $5 \cdot 3^x$ from $x = 0$ to $x = 2$?

A. 15

B. 20

C. 40

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q3/5

1634

QUESTION ID

S53-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20

GRADING NOTE $(45 - 5) \div 2 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q3/5

True or false?

An exponential growth function has a larger average rate of change on later intervals of the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q4/5

1635

QUESTION ID
S53-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its outputs, and their changes, keep multiplying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q4/5

True or false?

The average rate of change of 2^x from $x = 0$ to $x = 3$ is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q5/5

1636

QUESTION ID

S53-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(8 - 1) \div 3 = \frac{7}{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSF-LE.A.1.a

FORWARD 1

Derive the formula for the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Let $S = 1 + 2 + 4 + \ldots + 2^9$. Show that $2S - S = 2^{10} - 1$, and find S.

S = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q1/5

1638

QUESTION ID

S53-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

1,023

GRADING NOTE REQUIRED WORK: 2S written as $2 + 4 + \ldots + 2^{10}$, then subtracting S so that every term except 2^{10} and 1 cancels.

Form required: the subtraction and the value of S.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $S = a(1 - r^n) \div (1 - r)$ to find the sum of the first 6 terms of 1, 3, 9, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q2/5

1639

QUESTION ID
S53-F1-Q2

TYPE
Short answer

CORRECT ANSWER
364

GRADING NOTE Any equivalent form. $(1 - 729) \div (1 - 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q2/5

When deriving the sum formula, why do we multiply S by r?

A. To double the sum

B. So that S − rS cancels all but two terms

C. To make every term equal

D. To make r equal 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q3/5

1640

QUESTION ID

S53-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. So that S − rS cancels all but two terms

GRADING NOTE S and rS share every term except the first term of S and the last term of rS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q3/5

True or false?

For $r \neq 1$, $a + ar + \ldots + ar^{n-1} = a(1 - r^n) \div (1 - r)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q4/5

1641

QUESTION ID

S53-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the geometric series formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1 + 3 + 9 + 27 = 3^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q5/5

1642

QUESTION ID

S53-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum is 40, and $3^4 = 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSF-LE.A.1.a

FORWARD 2

Use logarithms to turn equal factors into equal differences • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 4 \cdot 2^x$.

Show that $\log_2 g(x) = x + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q1/5

1644

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\log_2(4 \cdot 2^x) = \log_2$

$4 + \log_2 2^x = 2 + x.$

GRADING NOTE Form required: the steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 10^x$.

What is log y?

log y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q2/5

1645

QUESTION ID

S53-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q2/5

The values y = 1, 10, 100, 1,000 occur at x = 0, 1, 2, 3.

What are the values of log y?

A. 0, 1, 2, 3

B. 0, 10, 20, 30

C. 1, 2, 3, 4

D. 1, 10, 100, 1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q3/5

1646

QUESTION ID

S53-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0, 1, 2, 3

GRADING NOTE $\log 10^k = k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q3/5

True or false?

Taking the logarithm of an exponential function $a \cdot b^x$ gives a linear function of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q4/5

1647

QUESTION ID

S53-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\log(a \cdot b^x) = \log a + x \log b$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q4/5

True or false?

$\log(100 \cdot 10^x) = 100x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q5/5

1648

QUESTION ID

S53-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log(100 \cdot 10^x) = 2 + x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q5/5

Recognize situations in which a quantity changes at a constant rate per unit interval

MAIN	Algebra I main question	**HSF-LE.A.1.b**
BACKWARD 1	Find the constant of proportionality	**7.RP.A.2.b**
BACKWARD 2	Write a linear function from a rate and an initial value	**8.F.B.4**
FORWARD 1	Relate arc length and sector area to the central angle	**HSG-C.B.5**
FORWARD 2	Use radian measure as arc length on a circle	**HSF-TF.A.1**

SET 54 • HSF-LE.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

A pool fills at a constant rate. At 2 p.m. it holds 1,500 gallons; at 5 p.m. it holds 3,300 gallons.

What is the rate, and how much water was in it at noon?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q1/5

1651

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

600 gallons per hour; 300 gallons at noon

GRADING NOTE Form required: the rate with units and the amount. 1,800 ÷ 3 = 600, and 1,500 − 2(600) = 300.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plant is 6 cm tall on day 0 and grows 0.8 cm each day.

How tall is it on day 15?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q2/5

1652

QUESTION ID

S54-M-Q2

TYPE

Short answer

CORRECT ANSWER

18 cm

GRADING NOTE Any equivalent form. 6 + 0.8(15).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A taxi charges the same amount for every mile.

Explain why the fare for the miles driven is a linear function of the miles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q3/5

1653

QUESTION ID
S54-M-Q3

TYPE
Short answer

CORRECT ANSWER
Each mile adds the same amount, so the fare grows by equal differences over equal intervals, which is linear growth.

GRADING NOTE Any equivalent explanation that uses the constant rate per mile.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which situation is modeled by a linear function?

A. A cell population doubles each hour.

B. A tank loses 4 liters each minute.

C. A loan grows 4% each month.

D. A car loses 15% of its value each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q4/5

1654

QUESTION ID

S54-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A tank loses 4 liters each minute.

GRADING NOTE Only the tank changes by the same amount each unit of time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q4/5

True or false?

A savings account that earns the same dollar amount of interest each year grows linearly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q5/5

1655

QUESTION ID

S54-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A constant amount per year is a constant rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q5/5

SET 54 • HSF-LE.A.1.b

BACKWARD 1

Find the constant of proportionality • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

ANSWER KEY Reduced copy of the question page, with its answer

y = 2.5x gives the cost of x pounds of apples.

What is the constant of proportionality?

Constant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q1/5

1657

QUESTION ID

S54-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A proportional table has x = 2, 4, 6 and y = 7, 14, 21.

What is the unit rate?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q2/5

1658

QUESTION ID

S54-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.5

GRADING NOTE Any equivalent form. 7 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A cyclist rides 12 miles in 3 hours at a constant rate.

What is the constant of proportionality in miles per hour?

A. 15

B. 36

C. $\frac{1}{4}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q3/5

1659

QUESTION ID

S54-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

GRADING NOTE 12 ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q3/5

True or false?

In y = 9x, y increases by 9 whenever x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q4/5

1660

QUESTION ID

S54-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 9 is the unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q4/5

True or false?

A proportional relationship through (5, 20) has constant of proportionality 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q5/5

1661

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $20 \div 5 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSF-LE.A.1.b

BACKWARD 2

Write a linear function from a rate and an initial value • Grade 8 • Functions • 8.F.B.4

A line passes through (2, 1,500) and (5, 3,300).

Find its slope and its y-intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q1/5

1663

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Slope 600;
y-intercept 300

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A phone battery is at 80% and drops 5 percentage points per hour.

Write a function B(t) for the charge after t hours.

B(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q2/5

1664

QUESTION ID

S54-B2-Q2

TYPE

Short answer

CORRECT ANSWER

80 − 5t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q2/5

Which function has rate of change 0.8 and initial value 6?

A. $y = 6x + 0.8$

B. $y = 0.8x + 6$

C. $y = 6.8x$

D. $y = 6(0.8)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q3/5

1665

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 0.8x + 6$

GRADING NOTE $y = mx + b$ with $m = 0.8$ and $b = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q3/5

True or false?

A function with initial value 10 and rate of change −2 is y = 10 − 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q4/5

1666

QUESTION ID

S54-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE b = 10 and m = −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q4/5

True or false?

The line through (0, 4) and (2, 10) has rate of change 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q5/5

1667

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (10 − 4) ÷ 2 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSF-LE.A.1.b

FORWARD 1

Relate arc length and sector area to the central angle • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

In a circle of radius 5 cm, by how much does the arc length increase for each 1-radian increase in the central angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q1/5

1669

QUESTION ID

S54-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5 cm

GRADING NOTE Any equivalent form. s = 5θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q1/5

A circle has radius 10.

Find the arc lengths for central angles of 0.5, 1, and 1.5 radians.

Arc lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q2/5

1670

QUESTION ID

S54-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5, 10, and 15

GRADING NOTE Form required: all three, in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q2/5

The area of a sector is $A = \frac{1}{2}r^2\theta$, and $r = 4$.

By how much does A increase when θ increases by 1 radian?

A. 4

B. 2

C. 16

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q3/5

1671

QUESTION ID

S54-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8

GRADING NOTE $\frac{1}{2}(16)(1) = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q3/5

True or false?

For a fixed radius, arc length is a linear function of the central angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q4/5

1672

QUESTION ID

S54-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $s = r\theta$ has the constant rate r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q4/5

True or false?

Doubling the radius while keeping the angle the same doubles the area of a sector.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q5/5

1673

QUESTION ID
S54-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The area depends on r^2, so it is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSF-LE.A.1.b

FORWARD 2

Use radian measure as arc length on a circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Convert 180° to radians.

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q1/5

1675

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q1/5

A central angle cuts off an arc of length 6 on a circle of radius 2.

What is its radian measure?

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q2/5

1676

QUESTION ID

S54-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 6 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q2/5

Convert $\frac{\pi}{3}$ radians to degrees.

A. 30°

B. 90°

C. 120°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q3/5

1677

QUESTION ID

S54-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60°

GRADING NOTE $\frac{\pi}{3} \times 180 \div \pi = 60$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q3/5

True or false?

One radian is the angle that cuts off an arc as long as the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q4/5

1678

QUESTION ID

S54-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

2π radians equals 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q5/5

1679

QUESTION ID
S54-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 2π radians equals 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q5/5

Recognize situations in which a quantity grows or decays by a constant percent rate per unit interval

MAIN	Algebra I main question	**HSF-LE.A.1.c**
BACKWARD 1	Read growth and decay rates from exponential functions	**HSF-IF.C.8.b**
BACKWARD 2	Solve percent problems	**7.RP.A.3**
FORWARD 1	Find times in percent-rate models with logarithms	**HSF-LE.A.4**
FORWARD 2	Use natural logarithms with continuous growth	**HSF-BF.B.5**

SET 55 • HSF-LE.A.1.c

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

A car worth $24,000 loses 15% of its value each year.

Write a function V(t), and find the value after 3 years to the nearest dollar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q1/5

1682

QUESTION ID

S55-M-Q1

TYPE

Short answer

CORRECT ANSWER

$V(t) = 24{,}000(0.85)^t$; $14,739

GRADING NOTE Form required: the function and the rounded value. $0.85^3 = 0.614125$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why a quantity that grows by 6% each year grows by equal factors over equal intervals.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q2/5

1683

QUESTION ID
S55-M-Q2

TYPE
Short answer

CORRECT ANSWER
Each year the quantity is multiplied by the same factor, 1.06, so every 1-year interval multiplies it by 1.06.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A town's population increases by 2.5% per year.

By what factor does it grow in 2 years? What is the percent increase over 2 years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q3/5

1684

QUESTION ID
S55-M-Q3

TYPE
Short answer

CORRECT ANSWER
1.050625; about 5.06%

GRADING NOTE Form required: the factor and the percent. $1.025^2 = 1.050625$, slightly more than 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q3/5

Which situation is exponential?

A. A worker earns $15 each hour.

B. A medicine dose decreases by 20% each hour.

C. A tank gains 3 liters each second.

D. A runner covers 200 meters each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q4/5

1685

QUESTION ID

S55-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A medicine dose decreases by 20% each hour.

GRADING NOTE Only the dose changes by a constant percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q4/5

True or false?

A 10% increase followed by a 10% decrease returns a quantity to its original value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q5/5

1686

QUESTION ID
S55-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 1.1 × 0.9 = 0.99, a 1% loss.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q5/5

SET 55 • HSF-LE.A.1.c

BACKWARD 1

Read growth and decay rates from exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

$V = 24{,}000(0.85)^t$.

What is the yearly percent change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q1/5

1688

QUESTION ID
S55-B1-Q1

TYPE
Short answer

CORRECT ANSWER
A 15% decrease per year

GRADING NOTE Any equivalent form. $0.85 = 1 - 0.15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$P = 500(1.025)^t$.

Is this growth or decay, and at what rate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q2/5

1689

QUESTION ID

S55-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Growth, at 2.5% per unit of t

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function shows 4% decay per year?

A. $y = 300(1.04)^t$

B. $y = 300(0.04)^t$

C. $y = 300 - 0.04t$

D. $y = 300(0.96)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q3/5

1690

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 300(0.96)^t$

GRADING NOTE $1 - 0.04 = 0.96$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 50(1.5)^t$ grows by 50% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q4/5

1691

QUESTION ID

S55-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1.5 = 1 + 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 80(0.75)^t$ decreases by 75% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q5/5

1692

QUESTION ID

S55-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It keeps 75%, so it decreases by 25%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSF-LE.A.1.c

BACKWARD 2

Solve percent problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A $24,000 car loses 15% of its value.

What is its new value?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q1/5

1694

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$20,400

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 8% sales tax on $45?

Tax: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q2/5

1695

QUESTION ID

S55-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$3.60

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the price of a $60 item after a 25% discount?

A. $15

B. $45

C. $75

D. $35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q3/5

1696

QUESTION ID

S55-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $45

GRADING NOTE 60 × 0.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

20% of 150 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q4/5

1697

QUESTION ID

S55-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 0.2 × 150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q4/5

True or false?

A 50% increase of 40 is 60, and a 50% decrease of 60 is 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q5/5

1698

QUESTION ID

S55-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 50% decrease of 60 is 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSF-LE.A.1.c

FORWARD 1

Find times in percent-rate models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$V(t) = 24{,}000(0.85)^t$.

When is V = 12,000? Round to the nearest tenth of a year.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q1/5

1700

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4.3

GRADING NOTE Form required: rounded to the nearest tenth. t = log 0.5 ÷ log 0.85 ≈ 4.27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $1.025^t = 2$. Round to the nearest tenth.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q2/5

1701

QUESTION ID

S55-F1-Q2

TYPE

Short answer

CORRECT ANSWER

28.1

GRADING NOTE Form required: rounded to the nearest tenth. log 2 ÷ log 1.025 ≈ 28.07.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which expression gives the half-life of $y = (0.9)^t$?

A. 0.9 ÷ 2

B. 0.5 ÷ 0.9

C. log 0.9 ÷ log 0.5

D. log 0.5 ÷ log 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q3/5

1702

QUESTION ID

S55-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. log 0.5 ÷ log 0.9

GRADING NOTE Solve $0.9^t = 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q3/5

True or false?

$\log(0.85^t) = t \log 0.85$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q4/5

1703

QUESTION ID

S55-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The power property of logarithms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q4**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q4/5

True or false?

If $2^t = 5$, then t = 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q5/5

1704

QUESTION ID

S55-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^{2.5} \approx 5.66$. t = log 5 ÷ log 2 ≈ 2.32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSF-LE.A.1.c

FORWARD 2

Use natural logarithms with continuous growth • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of $y = e^x$.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q1/5

1706

QUESTION ID
S55-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$y = \ln x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $e^{0.05t} = 3$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q2/5

1707

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

20 ln 3

GRADING NOTE Form required: exact. 0.05t = ln 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is $\ln(e^4)$?

A. 4e

B. 4

C. 1

D. e

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q3/5

1708

QUESTION ID

S55-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE In and e^x are inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q3/5

True or false?

e raised to the power ln 7 equals 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q4/5

1709

QUESTION ID

S55-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE e^x and ln x undo each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\ln 1 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q5/5

1710

QUESTION ID

S55-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $e^0 = 1$, so $\ln 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q5/5

Construct linear and exponential functions from a graph, a description, two points, or a table

MAIN	Algebra I main question	**HSF-LE.A.2**
BACKWARD 1	Write a linear function from two points or a table	**8.F.B.4**
BACKWARD 2	Write formulas for arithmetic and geometric sequences	**HSF-BF.A.2**
FORWARD 1	Build a sinusoidal model from a description	**HSF-TF.B.5**
FORWARD 2	Find a quadratic through three points with a system	**HSA-REI.C.8**

SET 56 • HSF-LE.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the linear function whose graph passes through (2, 11) and (6, 23).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q1/5

1713

QUESTION ID

S56-M-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 5

GRADING NOTE Any equivalent form. The slope is 12 ÷ 4 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the exponential function $y = ab^x$ $(b > 0)$ whose graph passes through (1, 6) and (3, 54).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q2/5

1714

QUESTION ID
S56-M-Q2

TYPE
Short answer

CORRECT ANSWER
$2(3)^x$

GRADING NOTE Any equivalent form. $ab^3 \div (ab) = b^2 = 9$, so $b = 3$ and $a = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

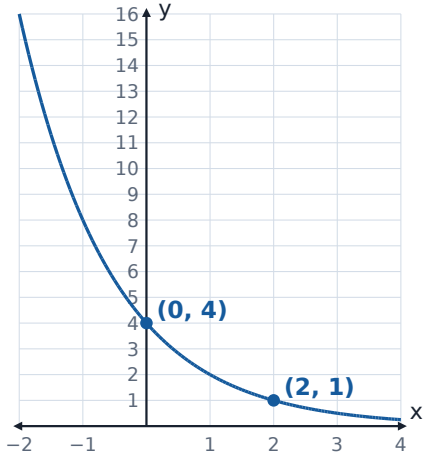
High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of an exponential function $y = ab^x$ passes through the marked points.

Write the function.

y = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q3/5

1715

QUESTION ID
S56-M-Q3

TYPE
Short answer

CORRECT ANSWER
$4(0.5)^x$

GRADING NOTE Any equivalent form, such as $4(1/2)^x$. $a = 4$, and $4b^2 = 1$ gives $b = 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 1.5, 3, 6, 12.

Which function fits?

A. $y = 2(1.5)^x$

B. $y = 1.5x + 1.5$

C. $y = 3x$

D. $y = 1.5(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q4/5

1716

QUESTION ID

S56-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 1.5(2)^x$

GRADING NOTE y starts at 1.5 and doubles each time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q4/5

True or false?

The linear function with f(0) = 10 that decreases by 2 for each unit increase in x is $f(x) = 10(0.8)^x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q5/5

1717

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal differences give f(x) = 10 − 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q5/5

SET 56 • HSF-LE.A.2

BACKWARD 1

Write a linear function from two points or a table • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Write the linear function through (0, −3) and (4, 5).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q1/5

1719

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2x − 3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 1, 2, 3 and y = 10, 7, 4.

Write the linear function.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q2/5

1720

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13 − 3x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q2/5

Which line passes through (−1, 2) and (1, 6)?

A. y = −2x + 4

B. y = 2x + 4

C. y = 2x + 2

D. y = 4x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q3/5

1721

QUESTION ID

S56-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 4

GRADING NOTE The slope is 4 ÷ 2 = 2; 6 = 2 + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q3/5

True or false?

The line through (0, 0) and (3, 6) is y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q4/5

1722

QUESTION ID

S56-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The slope is 2 and the y-intercept is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q4/5

True or false?

The line through (1, 1) and (2, 3) is y = 2x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q5/5

1723

QUESTION ID

S56-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE y = 2x − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSF-LE.A.2

BACKWARD 2

Write formulas for arithmetic and geometric sequences • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A geometric sequence has a(1) = 2 and ratio 3.

Write an explicit formula.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q1/5

1725

QUESTION ID

S56-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$2 \cdot 3^{n-1}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an explicit formula for 10, 7, 4, ...

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q2/5

1726

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

13 − 3n

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a recursive formula for 6, 3, 1.5, ...?

A. $a(n) = 6 - 3n$

B. $a(1) = 6,\ a(n) = 0.5a(n - 1)$

C. $a(1) = 6,\ a(n) = 2a(n - 1)$

D. $a(1) = 6,\ a(n) = a(n - 1) - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q3/5

1727

QUESTION ID

S56-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $a(1) = 6,\ a(n) = 0.5a(n - 1)$

GRADING NOTE Each term is half the one before.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q3/5

True or false?

An arithmetic sequence is a linear function on the positive integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q4/5

1728

QUESTION ID

S56-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal differences between consecutive terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q4/5

True or false?

The geometric sequence 5, 10, 20, ... has ratio 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q5/5

1729

QUESTION ID
S56-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The ratio is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSF-LE.A.2

FORWARD 1

Build a sinusoidal model from a description • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A tide has high water 10 ft and low water 2 ft, with period 12 hours and high tide at t = 0.

Write a cosine model h(t).

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q1/5

1731

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4cos(πt ÷ 6) + 6

GRADING NOTE Any equivalent form. Amplitude 4, midline 6, and 2π ÷ 12 = π ÷ 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sinusoid has maximum 10 and minimum 2.

What are its amplitude and midline?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q2/5

1732

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Amplitude 4; midline y = 6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q2/5

A model y = cos(Bt) has period 12.

What is B?

A. 12

B. π ÷ 12

C. π ÷ 6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q3/5

1733

QUESTION ID

S56-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. π ÷ 6

GRADING NOTE 2π ÷ B = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q3/5

True or false?

A sinusoidal model with maximum 8 and minimum 0 has amplitude 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q4/5

1734

QUESTION ID
S56-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE (8 − 0) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q4/5

True or false?

y = 3cos(2t) has period 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q5/5

1735

QUESTION ID

S56-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The period is 2π ÷ 2 = π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSF-LE.A.2

FORWARD 2

Find a quadratic through three points with a system • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

ANSWER KEY Reduced copy of the question page, with its answer

$y = ax^2 + bx + c$ passes through (0, 1), (1, 2), and (2, 7).

Write the system of equations for a, b, and c.

System: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q1/5

1737

QUESTION ID

S56-F2-Q1

TYPE

Short answer

CORRECT ANSWER

c = 1; a + b + c = 2; 4a + 2b + c = 7

GRADING NOTE Any equivalent form. Substitute each point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q1**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve that system.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q2/5

1738

QUESTION ID

S56-F2-Q2

TYPE

Short answer

CORRECT ANSWER

a = 2, b = −1, c = 1

GRADING NOTE Form required: all three values. a + b = 1 and 4a + 2b = 6, so 2a = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q2**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q2/5

Which row of the augmented matrix comes from the point (2, 7)?

A. [4 2 1 | 7]

B. [2 4 1 | 7]

C. [2 2 1 | 7]

D. [4 2 7 | 1]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q3/5

1739

QUESTION ID

S56-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. [4 2 1 | 7]

GRADING NOTE 4a + 2b + c = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q3/5

True or false?

Three points with different x-values that are not on one line determine exactly one parabola $y = ax^2 + bx + c$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q4/5

1740

QUESTION ID

S56-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The 3-by-3 system has exactly one solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q4/5

True or false?

The coefficient matrix of that system has 2 rows.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q5/5

1741

QUESTION ID
S56-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE There are 3 equations, so it has 3 rows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q5**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q5/5

Observe that exponential growth eventually exceeds linear, quadratic, and polynomial growth

MAIN	Algebra I main question	**HSF-LE.A.3**
BACKWARD 1	Tell equal differences from equal factors	**HSF-LE.A.1.a**
BACKWARD 2	Compare two functions given in different forms	**8.F.A.2**
FORWARD 1	Compare the growth of polynomial functions	**HSF-IF.C.7.c**
FORWARD 2	Compare logarithmic growth with other growth	**HSF-BF.B.5**

SET 57 • HSF-LE.A.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 100x$ and $g(x) = 2^x$.

What is the smallest whole number $x \geq 1$ for which $g(x) > f(x)$?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q1/5

1744

QUESTION ID

S57-M-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $g(9) = 512 < 900 = f(9)$, and $g(10) = 1{,}024 > 1{,}000 = f(10)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = 1.5^x$.

Which is larger at x = 10? At x = 20? What happens for larger x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q2/5

1745

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

f at x = 10 (100 compared with about 57.7); g at x = 20 (about 3,325 compared with 400); g stays larger from then on.

GRADING NOTE Form required: all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Plan A pays $1,000 per day. Plan B pays 1 cent on day 1, and its daily pay doubles each day.

On which day does Plan B's daily pay first exceed Plan A's?

Day: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q3/5

1746

QUESTION ID

S57-M-Q3

TYPE

Short answer

CORRECT ANSWER

Day 18

GRADING NOTE Any equivalent form. Day n pays $0.01 \cdot 2^{n-1}$ dollars: day 17 pays $655.36 and day 18 pays $1,310.72.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q3/5

Which function eventually has the greatest values as x increases?

A. $y = 50x^2$

B. $y = 1.01^x$

C. $y = x^3 + 500$

D. $y = 1{,}000x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q4/5

1747

QUESTION ID

S57-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 1.01^x$

GRADING NOTE Any exponential function with base greater than 1 eventually exceeds every polynomial function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-M-Q4**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q4/5

True or false?

Because $3x^2 > 2^x$ at $x = 5$, $3x^2$ stays greater than 2^x for all $x > 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q5/5

1748

QUESTION ID

S57-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At $x = 10$, $3x^2 = 300$ and $2^x = 1{,}024$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q5/5

SET 57 • HSF-LE.A.3

BACKWARD 1

Tell equal differences from equal factors • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 3, 6, 12, 24.

Is it linear or exponential? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q1/5

1750

QUESTION ID

S57-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Exponential: y is multiplied by 2 each time.

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 3, 6, 9, 12.

Is it linear or exponential? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q2/5

1751

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Linear: y increases by 3 each time.

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function grows by equal factors over equal intervals?

A. $y = 7 + 1.1x$

B. $y = 7(1.1)^x$

C. $y = 7x$

D. $y = 1.1x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q3/5

1752

QUESTION ID
S57-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. $y = 7(1.1)^x$

GRADING NOTE Only the exponential function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q3/5

True or false?

Linear functions grow by equal differences over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q4/5

1753

QUESTION ID

S57-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The difference is the slope times the interval length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q4/5

True or false?

$y = 5x^2$ grows by equal factors over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S57-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q5/5

1754

QUESTION ID
S57-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $5(2)^2 \div 5(1)^2 = 4$ but $5(3)^2 \div 5(2)^2 = 2.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S57-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSF-LE.A.3

BACKWARD 2

Compare two functions given in different forms • Grade 8 • Functions • 8.F.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Function A is y = 4x + 1. Function B has x = 0, 1, 2 and y = 3, 5, 7.

Which has the greater rate of change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q1/5

1756

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

A: 4 compared with 2.

GRADING NOTE Form required: the function and both rates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q1/5

Function A is y = −x + 6. Function B is the line through (0, 2) and (2, 8).

Which has the greater y-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q2/5

1757

QUESTION ID

S57-B2-Q2

TYPE

Short answer

CORRECT ANSWER

A: 6 compared with 2.

GRADING NOTE Form required: the function and both intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q2/5

$f(x) = 3x - 2$. Function g has $g(0) = 1$ and $g(1) = 5$, and it is linear.

Which statement is true?

A. f has the greater rate of change.

B. f has the greater initial value.

C. g has the greater rate of change and the greater initial value.

D. f and g have the same rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q3/5

1758

GRADING NOTE g has rate 4 and initial value 1; f has rate 3 and initial value −2.

QUESTION ID

S57-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. g has the greater rate of change and the greater initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q3/5

True or false?

A linear function with slope 5 increases faster than one with slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q4/5

1759

QUESTION ID

S57-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It gains 5 per unit instead of 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = 2x + 10 has greater values than y = 3x for every x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q5/5

1760

QUESTION ID

S57-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For x > 10, 3x is greater.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSF-LE.A.3

FORWARD 1

Compare the growth of polynomial functions • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

For which $x > 0$ is $x^3 > 100x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q1/5

1762

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x > 100

GRADING NOTE Form required: an inequality. Divide by $x^2 > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which grows faster for large x, x^4 or $1{,}000x^3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q2/5

1763

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x^4: $x^4 \div (1{,}000x^3) = x \div 1{,}000$, which is greater than 1 when $x > 1{,}000$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

As $x \to \infty$, what does $f(x) = x^4 - 50x^3$ do?

A. $f(x) \to -\infty$

B. $f(x) \to -50$

C. $f(x) \to \infty$

D. $f(x) \to 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q3/5

1764

QUESTION ID

S57-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $f(x) \to \infty$

GRADING NOTE The leading term x^4 decides the end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q3/5

True or false?

A polynomial of higher degree with a positive leading coefficient eventually exceeds any polynomial of lower degree.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q4/5

1765

QUESTION ID
S57-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The ratio grows without bound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q4/5

True or false?

For every large x, $0.001x^3$ is less than $1{,}000x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q5/5

1766

QUESTION ID

S57-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For $x > 1{,}000{,}000$, $0.001x^3$ is greater.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSF-LE.A.3

FORWARD 2

Compare logarithmic growth with other growth • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find $\log_2 1{,}024$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q1/5

1768

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $2^{10} = 1{,}024$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

By what factor must x be multiplied for log x to increase by 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q2/5

1769

QUESTION ID
S57-F2-Q2

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE Any equivalent form. log(10x) = 1 + log x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q2/5

Which is greatest when x = 1,000,000?

A. $\sqrt{x}$

B. $\log_2 x$

C. ln x

D. log x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q3/5

1770

QUESTION ID

S57-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $\sqrt{x}$

GRADING NOTE $\sqrt{x} = 1{,}000$; $\log x = 6$; $\ln x \approx 13.8$; $\log_2 x \approx 19.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q3/5

True or false?

Eventually, log x grows more slowly than any linear function with positive slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q4/5

1771

QUESTION ID

S57-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE log x is the inverse of the fast-growing 10^x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q4/5

True or false?

$\log_2 x = 100$ when $x = 200$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q5/5

1772

QUESTION ID

S57-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 2^{100}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q5/5

Interpret the parameters of a linear or exponential function in context

MAIN	Algebra I main question	**HSF-LE.B.5**
BACKWARD 1	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**
BACKWARD 2	Interpret the rate and initial value of a linear function	**8.F.B.4**
FORWARD 1	Interpret the parameters of a sinusoidal model	**HSF-TF.B.5**
FORWARD 2	Interpret the mean and standard deviation of a normal model	**HSS-ID.A.4**

SET 58 • HSF-LE.B.5

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

C(m) = 0.35m + 40 is the cost in dollars of renting a car for one day and driving m miles.

What do 0.35 and 40 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q1/5

1775

QUESTION ID

S58-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.35: the cost per mile ($0.35); 40: the daily fee ($40) charged even for 0 miles.

GRADING NOTE Form required: both meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q1/5

$P(t) = 1{,}200(1.04)^t$ is a town's population t years after 2020.

What do 1,200 and 1.04 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q2/5

1776

QUESTION ID
S58-M-Q2

TYPE
Short answer

CORRECT ANSWER
1,200: the population in 2020; 1.04: the population grows 4% each year.

GRADING NOTE Form required: both meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$A(t) = 80(0.5)^{t/6}$ is the milligrams of a medicine left in the body after t hours.

What do 80, 0.5, and 6 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q3/5

1777

QUESTION ID

S58-M-Q3

TYPE

Short answer

CORRECT ANSWER

80 mg is the starting dose; half of the medicine remains after every 6 hours (a half-life of 6 hours).

GRADING NOTE Form required: all three meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$V(t) = 18{,}000 - 1{,}500t$ is a machine's value in dollars after t years.

What does $-1{,}500$ mean?

A. The machine loses $1,500 of value each year.

B. The machine loses 1,500% of its value each year.

C. The machine was bought for $1,500.

D. The machine is worth $18,000 after 1,500 years.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q4/5

1778

QUESTION ID
S58-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. The machine loses $1,500 of value each year.

GRADING NOTE The slope is the change in value per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q4/5

True or false?

In $y = 250(0.92)^t$, the value decreases by 92% each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q5/5

1779

QUESTION ID
S58-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 92% remains each year, so it decreases by 8%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q5/5

SET 58 • HSF-LE.B.5

BACKWARD 1

Interpret the parts of an expression in context • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A plumber charges 75 + 60h dollars for h hours of work.

What does 75 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q1/5

1781

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A $75 fee charged for every visit, before any hours of work.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In $3{,}000(1.05)^t$, what does the factor 1.05 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q2/5

1782

QUESTION ID
S58-B1-Q2

TYPE
Short answer

CORRECT ANSWER
The amount grows 5% for each unit of t.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q2/5

A box with a square base of side s and height h has surface area $2s^2 + 4sh$.

What does 4sh represent?

A. The volume

B. The area of the four side faces

C. The perimeter of the base

D. The area of the top and bottom

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q3/5

1783

QUESTION ID

S58-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The area of the four side faces

GRADING NOTE Each side face is s by h, and there are four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q3/5

True or false?

In 500 − 20w, the term −20w shows a decrease of 20 for each unit of w.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q4/5

1784

QUESTION ID

S58-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The coefficient −20 is the change per unit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q4/5

True or false?

In 1.5x + 7, the coefficient of x is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q5/5

1785

QUESTION ID

S58-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The coefficient of x is 1.5; 7 is the constant term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSF-LE.B.5

BACKWARD 2

Interpret the rate and initial value of a linear function • Grade 8 • Functions • 8.F.B.4

Cost y = 12x + 30 for x hours of bike rental.

What is the initial value, and what does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q1/5

1787

GRADING NOTE Form required: the value and its meaning.

QUESTION ID

S58-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30: a $30 fee before any hours are counted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q1/5

The water in a tank is y = 500 − 25t gallons after t minutes.

What does −25 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q2/5

1788

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The tank loses 25 gallons each minute.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q2/5

A plant's height is a linear function of time. It is 8 cm on day 0 and 20 cm on day 4.

What is the rate of change?

A. 5 cm per day

B. 20 cm per day

C. 8 cm per day

D. 3 cm per day

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q3/5

1789

QUESTION ID

S58-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 cm per day

GRADING NOTE (20 − 8) ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q3/5

True or false?

In y = 40x + 100, 100 is the value of y when x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q4/5

1790

QUESTION ID

S58-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q4/5

True or false?

In d = 65t, 65 is the starting distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q5/5

1791

QUESTION ID

S58-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 65 is the rate (distance per unit of time); the starting distance is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSF-LE.B.5

FORWARD 1

Interpret the parameters of a sinusoidal model • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

$T(t) = 15 + 10\sin(\pi t \div 12)$ is the temperature in °C t hours after 8 a.m.

What do 15 and 10 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q1/5

1793

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15 °C is the average (midline) temperature; the temperature rises and falls 10 °C above and below it.

GRADING NOTE Form required: both meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the period of T, and what does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q2/5

1794

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

24 hours: the temperature pattern repeats every day.

GRADING NOTE Form required: the period and its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q2/5

$h(t) = 1.5\cos(\pi t \div 6) + 3$ is the depth of water in meters.

What is the greatest depth?

A. 4.5 m

B. 3 m

C. 6 m

D. 1.5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q3/5

1795

QUESTION ID

S58-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4.5 m

GRADING NOTE 3 + 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q3/5

True or false?

In y = A sin(Bx) + D, D gives the midline.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q4/5

1796

QUESTION ID

S58-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph is centered on y = D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q4/5

True or false?

The amplitude of $y = 1.5\cos(\pi t \div 6) + 3$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q5/5

1797

QUESTION ID
S58-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The amplitude is 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSF-LE.B.5

FORWARD 2

Interpret the mean and standard deviation of a normal model • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Battery life is normal with mean 400 hours and standard deviation 20 hours.

What does the 20 tell you?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q1/5

1799

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A typical battery's life is about 20 hours from the mean.

GRADING NOTE Any equivalent wording about spread around the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

With the same model, about what percent of batteries last between 360 and 440 hours?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q2/5

1800

QUESTION ID
S58-F2-Q2

TYPE
Short answer

CORRECT ANSWER
About 95%

GRADING NOTE Any equivalent form. That is within 2 standard deviations of the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which change makes a normal curve wider?

A. Increasing the standard deviation

B. Increasing the mean

C. Decreasing the standard deviation

D. Decreasing the mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q3/5

1801

QUESTION ID
S58-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. Increasing the standard deviation

GRADING NOTE The standard deviation measures spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q3/5

True or false?

Changing the mean of a normal model shifts the curve left or right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q4/5

1802

QUESTION ID

S58-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The mean is its center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q4/5

True or false?

In a normal model with mean 50 and standard deviation 5, the value 60 is 10 standard deviations above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q5/5

1803

QUESTION ID

S58-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(60 - 50) \div 5 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q5/5

SET 59 • HSS-ID.A.1 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Represent data with dot plots, histograms, and box plots

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Display numerical data in plots	**6.SP.B.4**
BACKWARD 2	Compute measures of center and variation	**6.SP.B.5.c**
FORWARD 1	Decide whether a normal model fits a distribution	**HSS-ID.A.4**
FORWARD 2	Decide whether results are consistent with a model by simulation	**HSS-IC.A.2**

SET 59 • HSS-ID.A.1

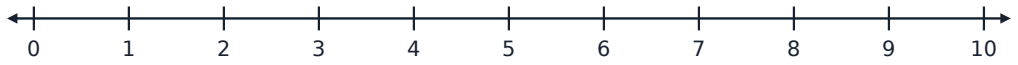
MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Make a dot plot of the numbers of books read by 10 students:

2, 3, 3, 4, 4, 4, 5, 5, 6, 9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q1/5

1806

QUESTION ID

S59-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

One dot above 2, two above 3, three above 4, two above 5, one above 6, and one above 9

GRADING NOTE Any correct dot plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q1/5

Find the five-number summary of 2, 3, 3, 4, 4, 4, 5, 5, 6, 9.

Summary: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q2/5

1807

QUESTION ID

S59-M-Q2

TYPE

Short answer

CORRECT ANSWER

Minimum 2, Q1 3, median 4, Q3 5, maximum 9

GRADING NOTE Form required: all five values. The median is the mean of the 5th and 6th values; Q1 and Q3 are the medians of the lower and upper five values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For the data 2, 3, 3, 4, 4, 4, 5, 5, 6, 9, use the 1.5 × IQR rule to decide whether 9 is an outlier.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q3/5

1808

QUESTION ID

S59-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes. IQR = 5 − 3 = 2, and Q3 + 1.5(2) = 8. Since 9 > 8, it is an outlier.

GRADING NOTE Form required: the answer with the fence.

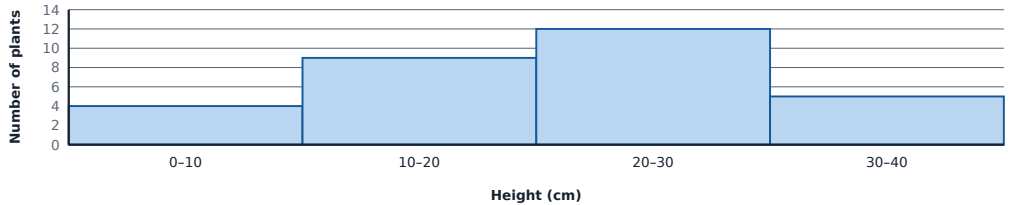
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows the heights of 30 plants. Each bar includes its lower edge but not its upper edge.

How many plants are at least 20 cm tall?



A. 21 **B.** 17 **C.** 5 **D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q4/5

1809

QUESTION ID

S59-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 17

GRADING NOTE The bars for 20–30 and 30–40 have 12 + 5 = 17 plants.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q4/5

True or false?

A box plot shows the mean of a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q5/5

1810

QUESTION ID
S59-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A box plot shows the five-number summary; the mean is not shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q5/5

SET 59 • HSS-ID.A.1

BACKWARD 1

Display numerical data in plots • Grade 6 • Statistics & Probability • 6.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Which display shows every data value, a dot plot or a box plot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q1/5

1812

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A dot plot

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Make a dot plot of 1, 2, 2, 3, 3, 3, 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q2/5

1813

GRADING NOTE Any correct dot plot.

QUESTION ID

S59-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

One dot above 1, two above 2, three above 3, and one above 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q2/5

Which display groups data values into intervals?

A. Box plot

B. Dot plot

C. Histogram

D. Ordered list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q3/5

1814

QUESTION ID

S59-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Histogram

GRADING NOTE A histogram counts the values in each interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q3/5

True or false?

A box plot is made from the five-number summary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q4/5

1815

QUESTION ID

S59-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Minimum, Q1, median, Q3, and maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q4/5

True or false?

A histogram shows each individual data value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q5/5

1816

QUESTION ID

S59-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It shows only how many values fall in each interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSS-ID.A.1

BACKWARD 2

Compute measures of center and variation • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the median of 3, 8, 5, 10, 4.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q1/5

1818

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. In order: 3, 4, 5, 8, 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 3, 8, 5, 10, 4.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q2/5

1819

QUESTION ID

S59-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 30 ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Q1 = 12 and Q3 = 20.

What is the interquartile range?

A. 32

B. 8

C. 4

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q3/5

1820

QUESTION ID

S59-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

GRADING NOTE 20 − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The range of 2, 7, 9, 15 is 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q4/5

1821

QUESTION ID

S59-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 15 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q4/5

True or false?

The median of 1, 2, 3, 100 is 26.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q5/5

1822

QUESTION ID

S59-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median is (2 + 3) ÷ 2 = 2.5; 26.5 is the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSS-ID.A.1

FORWARD 1

Decide whether a normal model fits a distribution • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A data set is strongly skewed to the right.

Is a normal model appropriate? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q1/5

1824

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No. A normal distribution is symmetric, and this one is not.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Scores are normal with mean 75 and standard deviation 8.

What score is 1 standard deviation above the mean?

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q2/5

1825

QUESTION ID
S59-F1-Q2

TYPE
Short answer

CORRECT ANSWER
83

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F1-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Values are normal with mean 20 and standard deviation 4.

About what percent are between 12 and 28?

A. 99.7%

B. 50%

C. 68%

D. 95%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q3/5

1826

QUESTION ID

S59-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 95%

GRADING NOTE 12 and 28 are 2 standard deviations from the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q3/5

True or false?

In a normal distribution, the mean and the median are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q4/5

1827

QUESTION ID

S59-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The distribution is symmetric about its mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

About 50% of a normal distribution lies within 1 standard deviation of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q5/5

1828

QUESTION ID
S59-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE About 68% does.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSS-ID.A.1

FORWARD 2

Decide whether results are consistent with a model by simulation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In 100 simulated sets of 10 fair coin flips, 8 or more heads happened in 5 sets.

Estimate the probability of 8 or more heads in 10 fair flips.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q1/5

1830

QUESTION ID
S59-F2-Q1

TYPE
Short answer

CORRECT ANSWER
0.05

GRADING NOTE Any equivalent form, such as 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner model says red comes up 25% of the time. In 200 spins, red came up 52 times.

What proportion is that? Is it consistent with the model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q2/5

1831

QUESTION ID
S59-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.26; yes, it is close to 0.25.

GRADING NOTE Form required: the proportion and a judgment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A model says a coin lands heads 50% of the time.

Which result most strongly questions the model?

A. 52 heads in 100 flips

B. 6 heads in 10 flips

C. 95 heads in 100 flips

D. 48 heads in 100 flips

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q3/5

1832

QUESTION ID

S59-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 95 heads in 100 flips

GRADING NOTE 95 of 100 is very far from 50 and almost never happens with a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A result that rarely happens in simulations of a model is evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q4/5

1833

QUESTION ID
S59-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE It suggests the model may be wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q4/5

True or false?

If a coin lands heads 6 times in 10 flips, the coin must be unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q5/5

1834

QUESTION ID

S59-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6 of 10 happens often with a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q5/5

SET 60 • HSS-ID.A.2 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Compare the center and spread of two or more data sets

MAIN	Algebra I main question	**HSS-ID.A.2**
BACKWARD 1	Tell measures of center from measures of variation	**6.SP.A.3**
BACKWARD 2	Compare two distributions using center and variability	**7.SP.B.3**
FORWARD 1	Use simulation to decide whether a difference between treatment…	**HSS-IC.B.5**
FORWARD 2	Evaluate reports based on data	**HSS-IC.B.6**

SET 60 • HSS-ID.A.2

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Class A scores: 70, 75, 80, 85, 90. Class B scores: 60, 70, 80, 90, 100.

Compare the centers and the spreads.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q1/5

1837

QUESTION ID

S60-M-Q1

TYPE

Short answer

CORRECT ANSWER

Both have mean and median 80. Class B is more spread out: its range is 40 compared with 20 (and its standard deviation is larger).

GRADING NOTE Form required: a comparison of center and of spread, with values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-M-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the population standard deviation (divide by n) of 2, 4, 4, 4, 5, 5, 7, 9.

Standard deviation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q2/5

1838

QUESTION ID

S60-M-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. The mean is 5; the squared deviations add to 32; 32 ÷ 8 = 4, and √4 = 2. (Dividing by n − 1 gives about 2.14.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q2/5

Store X's daily sales have median 50 and IQR 10. Store Y's have median 45 and IQR 25.

Which store typically sells more? Which is more consistent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q3/5

1839

QUESTION ID

S60-M-Q3

TYPE

Short answer

CORRECT ANSWER

Store X on both: its median is higher, and its IQR is smaller.

GRADING NOTE Form required: both answers with reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q3/5

Which measure of spread is best for skewed data with outliers?

A. Range

B. Standard deviation

C. Mean

D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q4/5

1840

QUESTION ID

S60-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Interquartile range

GRADING NOTE The IQR uses the middle half of the data, so outliers barely affect it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q4/5

True or false?

Two data sets with the same mean must have the same standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q5/5

1841

QUESTION ID

S60-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Classes A and B above have the same mean but different spreads.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q5/5

SET 60 • HSS-ID.A.2

BACKWARD 1

Tell measures of center from measures of variation • Grade 6 • Statistics & Probability • 6.SP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Which describes a typical value of a data set, the median or the IQR?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q1/5

1843

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The median

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of the data set 4, 4, 4, 4?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q2/5

1844

QUESTION ID

S60-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form. There is no variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q2/5

Which is a measure of variation?

A. Median

B. Mode

C. Mean

D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q3/5

1845

QUESTION ID

S60-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Interquartile range

GRADING NOTE The others are measures of center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q3/5

True or false?

A measure of center summarizes all of the values with a single number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q4/5

1846

QUESTION ID

S60-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, the mean or the median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q4/5

True or false?

The range is a measure of center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q5/5

1847

QUESTION ID

S60-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The range measures variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSS-ID.A.2

BACKWARD 2

Compare two distributions using center and variability • Grade 7 • Statistics & Probability • 7.SP.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Two classes have mean scores 78 and 84, and each has MAD 3.

How many MADs apart are the means?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q1/5

1849

QUESTION ID

S60-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. (84 − 78) ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Team A's mean height is 170 cm and Team B's is 180 cm; each has MAD 5 cm.

Express the difference of the means as a multiple of the MAD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q2/5

1850

QUESTION ID

S60-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2 MADs

GRADING NOTE Any equivalent form. 10 ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q2/5

Two means differ by 12, and each data set has MAD 4.

How many MADs is the difference?

A. 48

B. 12

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q3/5

1851

GRADING NOTE 12 ÷ 4.

QUESTION ID

S60-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q3/5

True or false?

When two distributions have little variability, a small difference in their centers is easier to notice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q4/5

1852

QUESTION ID

S60-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The data overlap less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q4/5

True or false?

If two data sets overlap, their means must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q5/5

1853

QUESTION ID

S60-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Overlapping sets can have different means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSS-ID.A.2

FORWARD 1

Use simulation to decide whether a difference between treatments is significant • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Plants given a fertilizer averaged 24 cm; the control group averaged 20 cm. In 1,000 random regroupings of the same plants, a difference of 4 cm or more happened 12 times.

Is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q1/5

1855

QUESTION ID
S60-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. A difference this large happened only about 1.2% of the time by random grouping alone, so chance is an unlikely explanation.

GRADING NOTE Form required: the answer with the simulated probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Why are subjects assigned to treatments at random in an experiment?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q2/5

1856

QUESTION ID
S60-F1-Q2

TYPE
Short answer

CORRECT ANSWER
So the groups are alike except for the treatment, and a difference can be credited to the treatment.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q2/5

In a regrouping simulation, a difference as large as the observed one happened 400 times out of 1,000.

What does that suggest?

A. The treatment definitely works.

B. The difference could easily be due to chance.

C. The treatment definitely does not work.

D. The experiment had no control group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q3/5

1857

QUESTION ID

S60-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The difference could easily be due to chance.

GRADING NOTE 40% is not rare.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q3/5

True or false?

A small simulated probability for the observed difference is evidence that the treatment had an effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q4/5

1858

QUESTION ID
S60-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Chance alone rarely produces it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q4/5

True or false?

In an experiment, the control group receives the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q5/5

1859

QUESTION ID
S60-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The control group does not receive the treatment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q5**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSS-ID.A.2

FORWARD 2

Evaluate reports based on data • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

An ad says "9 out of 10 dentists recommend Brand X," based on 10 dentists chosen by Brand X.

What is the flaw?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q1/5

1861

QUESTION ID
S60-F2-Q1

TYPE
Short answer

CORRECT ANSWER
The sample is tiny and was chosen by the company, so it is likely biased.

GRADING NOTE Any equivalent explanation that names size or bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An online poll of a website's visitors finds 80% support a policy.

Can the result be applied to all adults? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q2/5

1862

QUESTION ID

S60-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No. The respondents chose themselves and are only that site's visitors, so they do not represent all adults.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which sample is most likely to represent a school's students?

A. The first 50 students to arrive

B. The 50 students in the chess club

C. A random sample of 50 students from the school roster

D. 50 volunteers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q3/5

1863

QUESTION ID
S60-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. A random sample of 50 students from the school roster

GRADING NOTE Only random selection gives every student the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q3/5

True or false?

A larger random sample generally gives a smaller margin of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q4/5

1864

QUESTION ID
S60-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Estimates from larger random samples vary less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q4/5

True or false?

A survey with many volunteers is free of bias.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q5/5

1865

QUESTION ID

S60-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volunteers may differ from the population, whatever their number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q5**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSS-ID.A.3 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Interpret differences in shape, center, and spread, accounting for outliers

MAIN	Algebra I main question	**HSS-ID.A.3**
BACKWARD 1	Describe a distribution by its center, spread, and shape	**6.SP.A.2**
BACKWARD 2	Choose measures of center and spread that fit the data	**6.SP.B.5.d**
FORWARD 1	Use z-scores to judge how unusual a value is	**HSS-ID.A.4**
FORWARD 2	Distinguish a population parameter from a sample statistic	**HSS-IC.A.1**

SET 61 • HSS-ID.A.3

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Salaries at a small company, in thousands of dollars: 40, 42, 45, 45, 48, 50, 250.

Find the mean and the median. Which better describes a typical salary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q1/5

1868

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

Mean ≈ 74.29 thousand; median 45 thousand. The median is better: the one salary of 250 pulls the mean up.

GRADING NOTE Form required: both values (the mean to the nearest hundredth or as 520/7) and the choice with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q1/5

If the salary of 250 is removed, what happens to the mean and to the median?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q2/5

1869

QUESTION ID

S61-M-Q2

TYPE

Short answer

CORRECT ANSWER

The mean drops from about 74.29 to 45; the median stays 45.

GRADING NOTE Form required: both changes. 270 ÷ 6 = 45, and the middle values are 45 and 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q2/5

A histogram of wait times has a long tail on the right.

Describe its shape and compare its mean and median.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q3/5

1870

QUESTION ID

S61-M-Q3

TYPE

Short answer

CORRECT ANSWER

Skewed right; the mean is greater than the median.

GRADING NOTE Form required: the shape and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which data set has the largest standard deviation?

A. 28, 29, 30, 31, 32

B. 25, 30, 30, 30, 35

C. 10, 20, 30, 40, 50

D. 30, 30, 30, 30, 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q4/5

1871

QUESTION ID

S61-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10, 20, 30, 40, 50

GRADING NOTE All four have mean 30; the first is spread farthest from it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q4/5

True or false?

An outlier affects the median more than the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q5/5

1872

QUESTION ID

S61-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The mean uses every value; the median depends only on the middle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q5/5

SET 61 • HSS-ID.A.3

BACKWARD 1

Describe a distribution by its center, spread, and shape • Grade 6 • Statistics & Probability • 6.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Data: 1, 2, 2, 3, 3, 3, 4, 4, 5.

Describe the shape of the distribution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q1/5

1874

QUESTION ID

S61-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Symmetric, centered at 3

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value in 2, 3, 3, 3, 4, 15 is unusual?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q2/5

1875

QUESTION ID

S61-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q2/5

A dot plot has most values on the left and a few values far to the right.

What is its shape?

A. Skewed right

B. Symmetric

C. Uniform

D. Skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q3/5

1876

QUESTION ID

S61-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Skewed right

GRADING NOTE The tail points to the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q3/5

True or false?

A distribution can be described by its center, its spread, and its shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q4/5

1877

QUESTION ID

S61-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE These are the three main features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q4/5

True or false?

A symmetric distribution must have an outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q5/5

1878

QUESTION ID

S61-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Many symmetric distributions have no outliers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSS-ID.A.3

BACKWARD 2

Choose measures of center and spread that fit the data • Grade 6 • Statistics & Probability • 6.SP.B.5.d

For data with an extreme outlier, should you report the mean or the median as the center? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q1/5

1880

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

The median, because one extreme value barely changes it.

GRADING NOTE Form required: the choice and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For symmetric data with no outliers, which measures of center and spread are usually reported?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q2/5

1881

QUESTION ID

S61-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The mean and the mean absolute deviation

GRADING NOTE Any equivalent form. (The mean and the standard deviation are also accepted.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Home prices in a town include one mansion.

Which measure best describes a typical price?

A. The mean

B. The median

C. The maximum

D. The range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q3/5

1882

QUESTION ID

S61-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The median

GRADING NOTE The mansion pulls the mean up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q3/5

True or false?

The median is not changed much by a single extreme value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q4/5

1883

QUESTION ID

S61-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It depends on the middle of the ordered data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q4/5

True or false?

The IQR changes greatly when the largest value is made even larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q5/5

1884

QUESTION ID

S61-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The IQR depends on Q1 and Q3, which do not change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSS-ID.A.3

FORWARD 1

Use z-scores to judge how unusual a value is • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A distribution has mean 50 and standard deviation 4.

What is the z-score of 58?

z = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

1886

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. (58 − 50) ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A score of 92 on a test with mean 80 and standard deviation 6, or a score of 85 on a test with mean 70 and standard deviation 10: which is better relative to its test?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

1887

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

92: its z-score is 2, compared with 1.5.

GRADING NOTE Form required: the score and both z-scores.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a normal distribution, which z-score is the most unusual?

A. −3.1

B. −0.5

C. 1.2

D. 2.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

1888

QUESTION ID

S61-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −3.1

GRADING NOTE It is the farthest from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

True or false?

A z-score of 0 means the value equals the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

1889

QUESTION ID

S61-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is 0 standard deviations from the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A z-score of −2 is above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

1890

QUESTION ID
S61-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Negative z-scores are below the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSS-ID.A.3

FORWARD 2

Distinguish a population parameter from a sample statistic • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

What is the difference between a population and a sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q1/5

1892

QUESTION ID
S61-F2-Q1

TYPE
Short answer

CORRECT ANSWER
A population is the whole group of interest; a sample is part of it that is actually studied.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 200 students has a mean of 7.1 hours of sleep.

What does 7.1 estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q2/5

1893

QUESTION ID
S61-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The mean hours of sleep of all the students in the population.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q2/5

A parameter describes

A. a graph

B. one individual

C. a population

D. a sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q3/5

1894

QUESTION ID

S61-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a population

GRADING NOTE A statistic describes a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q3/5

True or false?

Different random samples from the same population can give different sample means.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q4/5

1895

QUESTION ID

S61-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is sampling variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q4/5

True or false?

A sample statistic always equals the population parameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q5/5

1896

QUESTION ID
S61-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is only an estimate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSS-ID.B.5 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Summarize categorical data in two-way frequency tables and interpret relative frequencies

MAIN	Algebra I main question	**HSS-ID.B.5**
BACKWARD 1	Read two-way tables for association	**8.SP.A.4**
BACKWARD 2	Write a part of a whole as a percent	**7.RP.A.3**
FORWARD 1	Use a two-way table to find conditional probabilities and test inde…	**HSS-CP.A.4**
FORWARD 2	Explain conditional probability and independence in everyday lang..	**HSS-CP.A.5**

SET 62 • HSS-ID.B.5

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows a survey of 200 students.

What percent of all the students play a sport?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q1/5

1899

QUESTION ID

S62-M-Q1

TYPE

Short answer

CORRECT ANSWER

52.5%

GRADING NOTE Any equivalent form. 105 ÷ 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What percent of Grade 10 students play a sport? What percent of Grade 9 students do?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q2/5

1900

QUESTION ID

S62-M-Q2

TYPE

Short answer

CORRECT ANSWER

60% of Grade 10; 45% of Grade 9

GRADING NOTE Form required: both conditional relative frequencies. 60 ÷ 100 and 45 ÷ 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Is there an association between grade and playing a sport? Explain.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q3/5

1901

QUESTION ID

S62-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes. 60% of Grade 10 students play a sport compared with 45% of Grade 9 students.

GRADING NOTE Form required: the answer and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the joint relative frequency of "Grade 9 and does not play a sport"?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. 50% **B.** 57.9% **C.** 55% **D.** 27.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q4/5

1902

QUESTION ID

S62-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 27.5%

GRADING NOTE 55 out of all 200 students. (55% is the share of Grade 9 students; about 57.9% is the share of non-players.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q4/5

True or false?

Of the students who play a sport, more than half are in Grade 10.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q5/5

1903

QUESTION ID

S62-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 60 ÷ 105 ≈ 57%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q5/5

SET 62 • HSS-ID.B.5

BACKWARD 1

Read two-way tables for association • Grade 8 • Statistics & Probability • 8.SP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

12 of 20 boys and 9 of 15 girls in a class like math.

How many students in all like math?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q1/5

1905

QUESTION ID

S62-B1-Q1

TYPE

Short answer

CORRECT ANSWER

21

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using the same class, is there an association between gender and liking math? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q2/5

1906

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. 12 ÷ 20 = 60% and 9 ÷ 15 = 60%: the same share.

GRADING NOTE Form required: the answer with both relative frequencies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What are the row and column totals of a two-way table called?

A. Joint frequencies

B. Conditional frequencies

C. Marginal frequencies

D. Outliers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q3/5

1907

QUESTION ID

S62-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Marginal frequencies

GRADING NOTE They are written in the margins of the table.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q3/5

True or false?

A two-way table shows data on two categorical variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q4/5

1908

QUESTION ID

S62-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One variable for the rows and one for the columns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a two-way table, every row must have the same total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q5/5

1909

QUESTION ID

S62-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Row totals can be different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSS-ID.B.5

BACKWARD 2

Write a part of a whole as a percent • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Write 45 out of 100 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q1/5

1911

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 60 out of 105 as a percent, rounded to the nearest tenth.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q2/5

1912

QUESTION ID

S62-B2-Q2

TYPE

Short answer

CORRECT ANSWER

57.1%

GRADING NOTE Form required: rounded to the nearest tenth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

27 is what percent of 60?

A. 45%

B. 60%

C. 33%

D. 27%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q3/5

1913

QUESTION ID

S62-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 45%

GRADING NOTE 27 ÷ 60 = 0.45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

55 out of 200 is 27.5%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q4/5

1914

QUESTION ID

S62-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 55 ÷ 200 = 0.275.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

40 out of 95 is 40%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q5/5

1915

QUESTION ID

S62-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 40 ÷ 95 ≈ 42.1%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSS-ID.B.5

FORWARD 1

Use a two-way table to find conditional probabilities and test independence • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A student from the table is chosen at random.

Find P(plays a sport | Grade 10).

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q1/5

1917

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form. 60 ÷ 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Are "Grade 10" and "plays a sport" independent? Explain with probabilities.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q2/5

1918

QUESTION ID
S62-F1-Q2

TYPE
Short answer

CORRECT ANSWER
No. P(plays a sport) = 0.525, but P(plays a sport | Grade 10) = 0.6.

GRADING NOTE Form required: the answer with both probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is P(Grade 9 | plays a sport)?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. About 0.43

B. 0.225

C. 0.525

D. 0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q3/5

1919

QUESTION ID
S62-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. About 0.43

GRADING NOTE 45 ÷ 105 ≈ 0.43.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Events A and B are independent if P(A | B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q4/5

1920

QUESTION ID

S62-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Knowing B does not change the probability of A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q4/5

True or false?

P(A | B) always equals P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q5/5

1921

QUESTION ID
S62-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE In the table, P(sport | Grade 10) = 0.6 but P(Grade 10 | sport) ≈ 0.57.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSS-ID.B.5

FORWARD 2

Explain conditional probability and independence in everyday language • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Explain in everyday language what P(late | rain) > P(late) means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q1/5

1923

QUESTION ID
S62-F2-Q1

TYPE
Short answer

CORRECT ANSWER
A person is more likely to be late on a rainy day than on a typical day.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A and B are independent, P(A) = 0.3, and P(B) = 0.5.

Find P(A and B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q2/5

1924

QUESTION ID
S62-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.15

GRADING NOTE Any equivalent form. 0.3 × 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of events is most likely independent?

A. Height and shoe size

B. Studying and passing a test

C. Flipping heads and rolling a 6

D. Rain and carrying an umbrella

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q3/5

1925

QUESTION ID
S62-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. Flipping heads and rolling a 6

GRADING NOTE A coin and a number cube do not affect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q3/5

True or false?

If A and B are independent, then P(A and B) = P(A) · P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q4/5

1926

QUESTION ID
S62-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the multiplication rule for independent events.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If P(A | B) = 0.8 and P(A) = 0.4, then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q5/5

1927

QUESTION ID
S62-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Knowing B doubles the probability of A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSS-ID.B.6.a • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Fit a linear, quadratic, or exponential function to data and use it to solve problems

MAIN	Algebra I main question	**HSS-ID.B.6.a**
BACKWARD 1	Describe patterns in scatter plots	**8.SP.A.1**
BACKWARD 2	Write linear and exponential functions through given points	**HSF-LE.A.2**
FORWARD 1	Use logarithms with an exponential model fitted to data	**HSF-LE.A.4**
FORWARD 2	Fit a sinusoidal model to periodic data	**HSF-TF.B.5**

SET 63 • HSS-ID.B.6.a

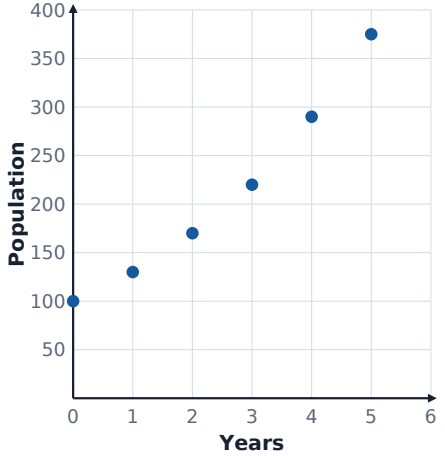
MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

ANSWER KEY Reduced copy of the question page, with its answer

The scatter plot shows a population over 5 years.

Which type of function fits best? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q1/5

1930

QUESTION ID

S63-M-Q1

TYPE

Short answer

CORRECT ANSWER

Exponential: the population is multiplied by about 1.3 each year (130 ÷ 100, 170 ÷ 130, ...).

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An exponential model for the data is $P(t) = 100(1.3)^t$.

Use it to predict the population at t = 8, to the nearest whole number.

P(8) ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q2/5

1931

QUESTION ID

S63-M-Q2

TYPE

Short answer

CORRECT ANSWER

816

GRADING NOTE Form required: rounded to the nearest whole number. $100(1.3)^8 \approx 815.7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q2/5

Data: x = 0, 1, 2, 3, 4 and y = 2, 5, 10, 17, 26.

Which type of function fits exactly? Write it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q3/5

1932

QUESTION ID

S63-M-Q3

TYPE

Short answer

CORRECT ANSWER

Quadratic: $y = x^2 + 1$

GRADING NOTE Form required: the type and the function. The second differences are all 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q3/5

A scatter plot rises and then falls in a symmetric curve.

Which model fits best?

A. Quadratic

B. Exponential growth

C. Exponential decay

D. Linear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q4/5

1933

QUESTION ID

S63-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Quadratic

GRADING NOTE A parabola rises and then falls symmetrically.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Using a model to predict far outside the range of the data is always reliable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q5/5

1934

QUESTION ID

S63-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Extrapolation assumes the pattern continues, which it may not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q5/5

SET 63 • HSS-ID.B.6.a

BACKWARD 1

Describe patterns in scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Points rise from left to right, close to a line.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

1936

QUESTION ID

S63-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A strong positive linear association

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is an outlier in a scatter plot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

1937

QUESTION ID

S63-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A point far from the overall pattern of the others

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

Points fall from left to right in a curved pattern.

Which describes the association?

A. Positive and nonlinear

B. Positive and linear

C. Negative and nonlinear

D. No association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

1938

QUESTION ID

S63-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Negative and nonlinear

GRADING NOTE Falling means negative; curved means nonlinear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A scatter plot shows the relationship between two quantitative variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

1939

QUESTION ID

S63-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each point is a pair of values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If the points show no pattern, there is a strong association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

1940

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No pattern means little or no association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSS-ID.B.6.a

BACKWARD 2

Write linear and exponential functions through given points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the exponential function through (0, 100) and (1, 130).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q1/5

1942

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$100(1.3)^x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the linear function through (0, 2) and (4, 26).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q2/5

1943

QUESTION ID

S63-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6x + 2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q2/5

Which exponential function $y = ab^x$ $(b > 0)$ passes through $(0, 50)$ and $(2, 12.5)$?

A. $y = 50(0.25)^x$

B. $y = 12.5(2)^x$

C. $y = 50(0.5)^x$

D. $y = 50 - 18.75x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q3/5

1944

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 50(0.5)^x$

GRADING NOTE $a = 50$ and $50b^2 = 12.5$, so $b = 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 3(2)^x$ passes through (2, 12).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q4/5

1945

QUESTION ID

S63-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3 \cdot 4 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q4/5

True or false?

$y = 5(1.2)^x$ passes through (0, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q5/5

1946

QUESTION ID

S63-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At x = 0, y = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSS-ID.B.6.a

FORWARD 1

Use logarithms with an exponential model fitted to data • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$P(t) = 100(1.3)^t$.

When does P reach 1,000? Round to the nearest tenth.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q1/5

1948

QUESTION ID

S63-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8.8

GRADING NOTE Form required: rounded to the nearest tenth. t = log 10 ÷ log 1.3 ≈ 8.78.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2(1.5)^x = 30$. Round to the nearest hundredth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q2/5

1949

QUESTION ID

S63-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6.68

GRADING NOTE Form required: rounded to the nearest hundredth. $1.5^x = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the exact solution of $100(1.3)^t = 500$?

A. $t = \log 1.3 \div \log 5$

B. $t = 500 \div 130$

C. $t = \log 5 \div \log 1.3$

D. $t = 5 \div 1.3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q3/5

1950

QUESTION ID

S63-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $t = \log 5 \div \log 1.3$

GRADING NOTE $1.3^t = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $1.3^t = 1.69$, then t = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q4/5

1951

QUESTION ID

S63-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1.3^2 = 1.69$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $10^t = 50$, then $t = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q5/5

1952

QUESTION ID

S63-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $t = \log 50 \approx 1.70$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSS-ID.B.6.a

FORWARD 2

Fit a sinusoidal model to periodic data • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Monthly average temperatures peak at 30 °C and bottom out at 10 °C.

What amplitude and midline fit a sinusoidal model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q1/5

1954

QUESTION ID

S63-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Amplitude 10; midline y = 20

GRADING NOTE Form required: both. (30 − 10) ÷ 2 and (30 + 10) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What period fits monthly temperatures that repeat every year?

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q2/5

1955

QUESTION ID
S63-F2-Q2

TYPE
Short answer

CORRECT ANSWER
12 months

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q2/5

Daylight hours vary from 9 to 15 over a year.

Which model fits?

A. A linear model

B. A sinusoidal model with amplitude 3 and period 12 months

C. A quadratic model with vertex at month 12

D. An exponential model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q3/5

1956

QUESTION ID

S63-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A sinusoidal model with amplitude 3 and period 12 months

GRADING NOTE The pattern repeats each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q3/5

True or false?

A repeating pattern in data suggests a sinusoidal model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q4/5

1957

QUESTION ID
S63-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Sinusoids are periodic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Data that vary between 9 and 15 have amplitude 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q5/5

1958

QUESTION ID
S63-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The amplitude is half the distance: 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q5/5

Assess the fit of a function by plotting and analyzing residuals

MAIN	Algebra I main question	**HSS-ID.B.6.b**
BACKWARD 1	Use a fitted function to make predictions	**HSS-ID.B.6.a**
BACKWARD 2	Subtract rational numbers	**7.NS.A.1.c**
FORWARD 1	Use logarithms to straighten exponential data	**HSF-LE.A.4**
FORWARD 2	Evaluate reports that use fitted models	**HSS-IC.B.6**

SET 64 • HSS-ID.B.6.b

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

ANSWER KEY Reduced copy of the question page, with its answer

The model $\hat{y} = 2.2x + 0.8$ is fitted to the data (1, 3), (2, 5), (3, 8), (4, 9), (5, 12).

Find the residual for each point.

Residuals: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q1/5

1961

QUESTION ID

S64-M-Q1

TYPE

Short answer

CORRECT ANSWER

0, −0.2, 0.6, −0.6, 0.2

GRADING NOTE Form required: all five, in order. Residual = actual − predicted; the predictions are 3, 5.2, 7.4, 9.6, 11.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q1/5

For that model, what does the residual −0.6 at x = 4 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q2/5

1962

QUESTION ID

S64-M-Q2

TYPE

Short answer

CORRECT ANSWER

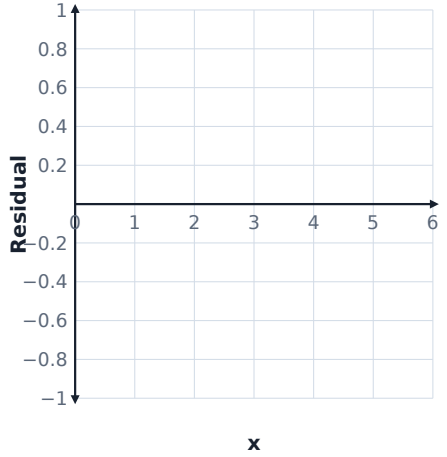
The actual y-value is 0.6 less than the model predicts.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q2/5

Make a residual plot for the residuals 0, −0.2, 0.6, −0.6, 0.2 at x = 1, 2, 3, 4, 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q3/5

1963

QUESTION ID

S64-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

The points (1, 0), (2, −0.2), (3, 0.6), (4, −0.6), and (5, 0.2), scattered above and below 0 with no pattern

GRADING NOTE Any correct plot. The lack of a pattern suggests the linear model fits well.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A residual plot shows a clear U-shaped pattern.

What does that suggest?

A. The correlation is exactly 1.

B. A linear model is not a good fit; a curved model may fit better.

C. The linear model fits well.

D. There are no residuals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q4/5

1964

QUESTION ID

S64-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A linear model is not a good fit; a curved model may fit better.

GRADING NOTE A pattern in the residuals means the model misses part of the relationship.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q4/5

True or false?

A good linear fit has residuals scattered randomly around 0, with no pattern.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q5/5

1965

QUESTION ID
S64-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The model then captures the pattern of the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q5/5

SET 64 • HSS-ID.B.6.b

BACKWARD 1

Use a fitted function to make predictions • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

ANSWER KEY Reduced copy of the question page, with its answer

The model is $\hat{y} = 2.2x + 0.8$.

Predict y when x = 6.

$\hat{y}$ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q1/5

1967

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE Any equivalent form. 13.2 + 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-B1-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The model is $y = 3x^2 - 2$.

Predict y when x = 4.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q2/5

1968

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

46

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-B1-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q2/5

The model is $y = 50(0.9)^x$.

Predict y when x = 2.

A. 50

B. 40.5

C. 45

D. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q3/5

1969

QUESTION ID

S64-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 40.5

GRADING NOTE 50(0.81) = 40.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-B1-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q3/5

True or false?

A predicted value comes from the model, not from the data table.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q4/5

1970

QUESTION ID

S64-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the model's output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q4/5

True or false?

If a model predicts 14 and the actual value is 14, the residual is 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q5/5

1971

QUESTION ID

S64-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The residual is 14 − 14 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSS-ID.B.6.b

BACKWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

Find 8 − 7.4.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q1/5

1973

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 9 − 9.6.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q2/5

1974

QUESTION ID

S64-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 5 − 5.2.

A. −10.2

B. 10.2

C. 0.2

D. −0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q3/5

1975

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −0.2

GRADING NOTE 5.2 is greater than 5, so the difference is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q3/5

True or false?

$3 - (-2) = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q4/5

1976

QUESTION ID

S64-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtracting −2 adds 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$-4 - 6 = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q5/5

1977

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-4 - 6 = -10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSS-ID.B.6.b

FORWARD 1

Use logarithms to straighten exponential data • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Data: $y = 1, 3, 9, 27$ at $x = 0, 1, 2, 3$.

Find $\log_3 y$ for each point.

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q1/5

1979

QUESTION ID

S64-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 1, 2, 3

GRADING NOTE Form required: all four, in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q1/5

A fitted line is log y = 0.3x + 1.

Write y as an exponential function of x.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q2/5

1980

QUESTION ID

S64-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$10 \cdot 10^{0.3x}$, which is about $10(2.0)^x$

GRADING NOTE Any equivalent form. $y = 10^{0.3x+1}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q2/5

Why take the logarithm of y for data that look exponential?

A. It removes all of the residuals.

B. It makes every y-value negative.

C. It doubles the data.

D. It makes the pattern linear, so a line can be fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q3/5

1981

QUESTION ID

S64-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. It makes the pattern linear, so a line can be fit.

GRADING NOTE $\log(ab^x)$ is linear in x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $y = 5 \cdot 2^x$, then log y is a linear function of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q4/5

1982

QUESTION ID

S64-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE log y = log 5 + x log 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log(5 \cdot 2^x) = 5 + 2x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q5/5

1983

QUESTION ID

S64-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log(5 \cdot 2^x) = \log 5 + x \log 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSS-ID.B.6.b

FORWARD 2

Evaluate reports that use fitted models • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A report fits a line to the heights of children aged 5 to 15 and uses it to predict the height of a 40-year-old.

What is the problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q1/5

1985

QUESTION ID
S64-F2-Q1

TYPE
Short answer

CORRECT ANSWER
It extrapolates far beyond the data; people stop growing, so the linear pattern does not continue.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q1/5

A report finds a strong correlation between ice cream sales and drownings and says ice cream causes drowning.

What is wrong?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q2/5

1986

QUESTION ID
S64-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Correlation does not show causation; hot weather increases both.

GRADING NOTE Any equivalent explanation that names a lurking variable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q2**
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q2/5

A residual plot shows a strong curve, but a report says the linear model is excellent.

What should you conclude?

A. The report is correct.

B. The data must be wrong.

C. The residuals are all 0.

D. The linear model may not be appropriate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q3/5

1987

QUESTION ID

S64-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The linear model may not be appropriate.

GRADING NOTE A curved residual pattern shows a poor linear fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q3/5

True or false?

A report should state how its data were collected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q4/5

1988

QUESTION ID

S64-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The method decides what conclusions are justified.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q4/5

True or false?

A model with a high correlation coefficient is always the right type of model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q5/5

1989

QUESTION ID
S64-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Curved data can have a high r for a line and still be fit better by a curve.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q5/5

SET 65 • HSS-ID.B.6.c • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Fit a linear function to a scatter plot that suggests a linear association

MAIN	Algebra I main question	**HSS-ID.B.6.c**
BACKWARD 1	Judge a line of fit informally	**8.SP.A.2**
BACKWARD 2	Write a line through two points	**HSF-LE.A.2**
FORWARD 1	Evaluate conclusions drawn from a line of fit	**HSS-IC.B.6**
FORWARD 2	Use simulation to judge whether an association could be due to ch…	**HSS-IC.B.5**

SET 65 • HSS-ID.B.6.c

MAIN

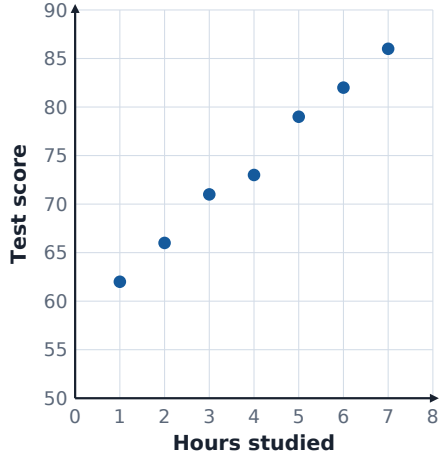
Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

ANSWER KEY Reduced copy of the question page, with its answer

The scatter plot shows hours studied and test scores.

Draw a line of fit and write its equation.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q1/5

1992

QUESTION ID

S65-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line close to the points, with about as many points above it as below it

Written: About y = 4x + 58

GRADING NOTE Any equation close to the least-squares line ŷ = 4x + 58.14 (slope about 3.5 to 4.5). The equation should match the student's line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The least-squares line for the data is $\hat{y} = 4x + 58.14$.

Predict the score for 4.5 hours, to the nearest whole number.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q2/5

1993

QUESTION ID

S65-M-Q2

TYPE

Short answer

CORRECT ANSWER

76

GRADING NOTE Form required: rounded to the nearest whole number. 18 + 58.14 = 76.14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through the data points (1, 62) and (7, 86), to use as a line of fit.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q3/5

1994

QUESTION ID

S65-M-Q3

TYPE

Short answer

CORRECT ANSWER

y = 4x + 58

GRADING NOTE Any equivalent form. The slope is 24 ÷ 6 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q3/5

Which line fits the points (0, 1), (1, 3), (2, 5), (3, 7) exactly?

A. y = 3x + 1

B. y = 2x + 1

C. y = 2x

D. y = x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q4/5

1995

QUESTION ID

S65-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 1

GRADING NOTE y increases by 2 for each 1 in x, starting at 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q4/5

True or false?

A line of fit should pass through every data point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q5/5

1996

QUESTION ID

S65-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It should be close to the points overall, not through all of them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q5/5

SET 65 • HSS-ID.B.6.c

BACKWARD 1

Judge a line of fit informally • Grade 8 • Statistics & Probability • 8.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Is a straight line a good model for points that curve strongly upward? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q1/5

1998

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

No. The points do not follow a straight pattern.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How can you judge whether a line of fit is good?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q2/5

1999

QUESTION ID

S65-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The points are close to the line, with about as many above it as below it.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A line of fit should have

A. all of the points below it

B. all of the points above it

C. about as many points above it as below it

D. exactly two points on it

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q3/5

2000

QUESTION ID
S65-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. about as many points above it as below it

GRADING NOTE It balances the points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q3/5

2000

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Lines of fit are used for data with a linear association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q4/5

2001

QUESTION ID

S65-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Straight-line patterns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A line of fit must pass through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q5/5

2002

QUESTION ID

S65-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its y-intercept can be any value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSS-ID.B.6.c

BACKWARD 2

Write a line through two points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through (2, 66) and (6, 82).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q1/5

2004

QUESTION ID

S65-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4x + 58

GRADING NOTE Any equivalent form. The slope is 16 ÷ 4 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through (0, 7) and (3, 13).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q2/5

2005

QUESTION ID

S65-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2x + 7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which line passes through (0, 10) and (5, 0)?

A. $y = 2x + 10$

B. $y = 10x - 2$

C. $y = -2x + 10$

D. $y = -0.5x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q3/5

2006

QUESTION ID

S65-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = -2x + 10$

GRADING NOTE The slope is $-10 \div 5 = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q3/5

True or false?

The line through (0, 3) with slope 4 is y = 4x + 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q4/5

2007

QUESTION ID

S65-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE y = mx + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The line through (2, 2) and (4, 8) has slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q5/5

2008

QUESTION ID
S65-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $6 \div 2 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSS-ID.B.6.c

FORWARD 1

Evaluate conclusions drawn from a line of fit • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

In a study of 7 students, each extra hour of study went with about 4 more points.

Can a school claim this holds for all of its students? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q1/5

2010

QUESTION ID
S65-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Not with confidence: the sample is small and was not chosen at random, and the study shows association, not cause.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report uses $\hat{y} = 4x + 58$ to predict a score of 138 for 20 hours of study.

Why is that unreasonable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q2/5

2011

QUESTION ID
S65-F1-Q2

TYPE
Short answer

CORRECT ANSWER
20 hours is far outside the data (1 to 7 hours), and scores cannot exceed 100.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which detail makes a regression conclusion more trustworthy?

A. The prediction is far outside the data.

B. The line was drawn by hand.

C. The data came from a large random sample.

D. The sample was 5 volunteers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q3/5

2012

QUESTION ID
S65-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. The data came from a large random sample.

GRADING NOTE Random, large samples represent the population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Observational data alone cannot prove that one variable causes another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q4/5

2013

QUESTION ID

S65-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Lurking variables may explain the association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A line of fit from 7 students proves the effect for every student.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q5/5

2014

QUESTION ID

S65-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It describes those 7 students only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSS-ID.B.6.c

FORWARD 2

Use simulation to judge whether an association could be due to chance • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

The test scores were randomly re-paired with the study hours 1,000 times, and the slope was never as large as 4.

What does that suggest?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

2016

QUESTION ID
S65-F2-Q1

TYPE
Short answer

CORRECT ANSWER
The association is very unlikely to be due to chance alone.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In an experiment, a tutoring group averaged 6 points more than a control group. Random regroupings gave differences of 6 or more 30% of the time.

What is the conclusion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

2017

QUESTION ID
S65-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The difference is not significant; it could easily be due to chance.

GRADING NOTE Form required: the conclusion with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the purpose of random assignment in an experiment?

A. To make the groups alike except for the treatment

B. To make the sample smaller

C. To guarantee a significant result

D. To remove the need for a control group

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

2018

QUESTION ID

S65-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. To make the groups alike except for the treatment

GRADING NOTE Then differences can be credited to the treatment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A result is statistically significant when it would rarely happen by chance alone.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

2019

QUESTION ID
S65-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE That is the meaning of significance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A difference that happens in 40% of random simulations is statistically significant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

2020

QUESTION ID
S65-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 40% is common, not rare.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

2020

SET 66 • HSS-ID.C.7 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Interpret the slope and the intercept of a linear model in the context of the data

MAIN	Algebra I main question	**HSS-ID.C.7**
BACKWARD 1	Interpret the parameters of linear and exponential functions	**HSF-LE.B.5**
BACKWARD 2	Use a linear model to solve problems	**8.SP.A.3**
FORWARD 1	Interpret the parameters of continuous exponential models	**HSF-LE.A.4**
FORWARD 2	Understand a fitted slope as an estimate of a population value	**HSS-IC.A.1**

SET 66 • HSS-ID.C.7

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

ANSWER KEY Reduced copy of the question page, with its answer

$\hat{y} = 4x + 58$ predicts a test score from x hours of study.

Interpret the slope and the intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q1/5

2023

QUESTION ID

S66-M-Q1

TYPE

Short answer

CORRECT ANSWER

Slope: each additional hour of study goes with about 4 more points (predicted). Intercept: the predicted score for 0 hours is 58.

GRADING NOTE Form required: both interpretations, in context.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q1/5

2023

ANSWER KEY Reduced copy of the question page, with its answer

A model for used cars is $\hat{y} = 22{,}000 - 1{,}800x$, where x is the age in years and $\hat{y}$ is the price in dollars.

Interpret $-1{,}800$ and $22{,}000$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q2/5

2024

QUESTION ID

S66-M-Q2

TYPE

Short answer

CORRECT ANSWER

The predicted price drops about $1,800 for each year of age; the predicted price of a new (0-year-old) car is $22,000.

GRADING NOTE Form required: both interpretations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q2/5

2024

ANSWER KEY Reduced copy of the question page, with its answer

$\hat{y} = 0.9x - 20$ predicts an adult's weight in pounds from height in inches.

Does the intercept make sense in context? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q3/5

2025

QUESTION ID
S66-M-Q3

TYPE
Short answer

CORRECT ANSWER
No. It predicts −20 pounds at a height of 0 inches, which is impossible and far outside the data.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q3/5

2025

$\hat{y} = 2.5x + 10$ predicts the cost in dollars of a pizza with x toppings.

What does 2.5 mean?

A. Each additional topping adds about $2.50 to the predicted cost.

B. A pizza has 2.5 toppings.

C. A pizza costs $2.50.

D. A plain pizza costs $2.50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q4/5

2026

QUESTION ID

S66-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Each additional topping adds about $2.50 to the predicted cost.

GRADING NOTE The slope is the change in predicted cost per topping.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q4/5

True or false?

The slope of a fitted line gives the exact change in y for every individual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q5/5

2027

QUESTION ID

S66-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives the predicted (average) change; individuals vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q5/5

SET 66 • HSS-ID.C.7

BACKWARD 1

Interpret the parameters of linear and exponential functions • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A technician charges C = 50 + 20h dollars for h hours.

What does 20 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q1/5

2029

QUESTION ID

S66-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$20 for each hour of work

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$P = 3{,}000(0.9)^t$ is a population after t years.

What does 0.9 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q2/5

2030

QUESTION ID

S66-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The population decreases 10% each year.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q2/5

$d = 300 - 60t$ is a driver's distance from home, in miles, after t hours.

What does 300 mean?

A. The starting distance from home

B. The speed

C. The distance driven each hour

D. The time to arrive home

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q3/5

2031

QUESTION ID
S66-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. The starting distance from home

GRADING NOTE At $t = 0$, $d = 300$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q3/5

True or false?

In y = 1.5x + 4, increasing x by 2 increases y by 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q4/5

2032

QUESTION ID
S66-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 1.5 × 2 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In $y = 8(1.25)^x$, y increases by 1.25 for each unit of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q5/5

2033

QUESTION ID
S66-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE y is multiplied by 1.25 for each unit of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSS-ID.C.7

BACKWARD 2

Use a linear model to solve problems • Grade 8 • Statistics & Probability • 8.SP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

y = 2x + 5 models a plant's height in cm after x weeks.

Predict the height at 6 weeks.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q1/5

2035

QUESTION ID

S66-B2-Q1

TYPE

Short answer

CORRECT ANSWER

17 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using y = 2x + 5, after how many weeks is the plant 21 cm tall?

Weeks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q2/5

2036

QUESTION ID

S66-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 2x + 5 = 21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In y = 2x + 5, what does 5 mean?

A. The height at week 0

B. The weeks needed to grow 1 cm

C. The final height

D. The growth per week

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q3/5

2037

QUESTION ID

S66-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The height at week 0

GRADING NOTE y = 5 when x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q3/5

True or false?

The slope of a linear model is a rate of change.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q4/5

2038

QUESTION ID

S66-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the change in y per unit of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q4/5

True or false?

In y = 2x + 5, the plant grows 5 cm each week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q5/5

2039

QUESTION ID

S66-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It grows 2 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSS-ID.C.7

FORWARD 1

Interpret the parameters of continuous exponential models • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$A(t) = 500e^{0.04t}$.

What does 0.04 mean? When does A double, to the nearest tenth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q1/5

2041

QUESTION ID

S66-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A continuous growth rate of 4% per unit of t; t ≈ 17.3

GRADING NOTE Form required: the meaning and the rounded time. t = ln 2 ÷ 0.04.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q1/5

$N(t) = 1{,}000e^{-0.1t}$.

What does −0.1 mean? When is N = 500, to the nearest tenth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q2/5

2042

QUESTION ID

S66-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Continuous decay at 10% per unit of t; t ≈ 6.9

GRADING NOTE Form required: the meaning and the rounded time. t = ln 2 ÷ 0.1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $e^{2x} = 9$ exactly.

A. $x = \ln 3$

B. $x = 4.5$

C. $x = e^{4.5}$

D. $x = \ln 9 \div 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q3/5

2043

QUESTION ID

S66-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = ln 3

GRADING NOTE $2x = \ln 9 = 2 \ln 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\ln(e^{0.04t}) = 0.04t$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q4/5

2044

QUESTION ID

S66-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ln and e^x undo each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $e^k = 1$, then $k = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q5/5

2045

QUESTION ID

S66-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $e^0 = 1$, so $k = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSS-ID.C.7

FORWARD 2

Understand a fitted slope as an estimate of a population value • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 50 used cars of one model gives $\hat{y} = 22{,}000 - 1{,}800x$.

What population value does $-1{,}800$ estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q1/5

2047

QUESTION ID
S66-F2-Q1

TYPE
Short answer

CORRECT ANSWER
The average yearly drop in price for all used cars of that model.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Would another random sample of 50 cars give exactly the same slope? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q2/5

2048

QUESTION ID
S66-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Probably not: different samples give different statistics (sampling variability).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q2/5

Which of these is a parameter?

A. The mean price of the 50 sampled cars

B. One car's price

C. The slope from the sample

D. The mean price of all used cars of this model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q3/5

2049

QUESTION ID
S66-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. The mean price of all used cars of this model

GRADING NOTE A parameter describes the whole population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Larger random samples tend to give statistics closer to the parameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q4/5

2050

QUESTION ID
S66-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Their sampling variability is smaller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q4/5

2050

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A statistic is a number that describes a whole population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q5/5

2051

QUESTION ID
S66-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A statistic describes a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q5/5

Compute (using technology) and interpret the correlation coefficient of a linear fit

MAIN	Algebra I main question	**HSS-ID.C.8**
BACKWARD 1	Describe the direction of an association	**8.SP.A.1**
BACKWARD 2	Fit and use a line for linear data	**HSS-ID.B.6.c**
FORWARD 1	Use simulation to judge whether a correlation could be due to ch…	**HSS-IC.B.5**
FORWARD 2	Evaluate claims based on a correlation	**HSS-IC.B.6**

SET 67 • HSS-ID.C.8

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Use technology to find the correlation coefficient r for the data

(1, 9), (2, 8), (3, 8), (4, 6), (5, 5), (6, 5), (7, 3), (8, 2),

to the nearest hundredth. Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q1/5

2054

GRADING NOTE Form required: r to the nearest hundredth and a description.

QUESTION ID

S67-M-Q1

TYPE

Short answer

CORRECT ANSWER

r ≈ −0.98: a strong negative linear association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

r = −0.85 for hours of TV and test score.

Interpret this value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q2/5

2055

QUESTION ID

S67-M-Q2

TYPE

Short answer

CORRECT ANSWER

A strong negative linear association: more TV goes with lower scores.

GRADING NOTE Any equivalent interpretation of the direction, strength, and form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value of r shows the strongest linear relationship?

A. 0.10

B. −0.92

C. −0.40

D. 0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q3/5

2056

QUESTION ID

S67-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −0.92

GRADING NOTE Strength depends on |r|, and |−0.92| is the largest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Can a correlation coefficient equal 1.3? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q4/5

2057

QUESTION ID

S67-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. r is always between −1 and 1.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q4/5

True or false?

r = 0 means there is no relationship of any kind between the variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q5/5

2058

QUESTION ID

S67-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means no linear relationship; there could be a curved one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q5/5

SET 67 • HSS-ID.C.8

BACKWARD 1

Describe the direction of an association • Grade 8 • Statistics & Probability • 8.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Points rise from left to right, fairly close to a line.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q1/5

2060

QUESTION ID

S67-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A positive linear association

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q1/5

2060

ANSWER KEY Reduced copy of the question page, with its answer

Points for hours of sleep and number of errors fall from left to right.

Is the association positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q2/5

2061

QUESTION ID

S67-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Negative

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of variables likely has a positive association?

A. Height and arm span

B. Outdoor temperature and heating cost

C. A car's age and its value

D. Speed and travel time for a fixed trip

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q3/5

2062

QUESTION ID

S67-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Height and arm span

GRADING NOTE Taller people tend to have longer arm spans.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q3/5

True or false?

In a negative association, as one variable increases, the other tends to decrease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q4/5

2063

QUESTION ID

S67-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the meaning of negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Clusters in a scatter plot mean the data are linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q5/5

2064

QUESTION ID

S67-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Clusters describe grouping, not form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSS-ID.C.8

BACKWARD 2

Fit and use a line for linear data • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

ANSWER KEY Reduced copy of the question page, with its answer

Write a line of fit through (2, 5) and (8, 14).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q1/5

2066

QUESTION ID

S67-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.5x + 2

GRADING NOTE Any equivalent form. The slope is 9 ÷ 6 = 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The least-squares line is $\hat{y} = 1.5x + 2$.

Predict y when $x = 6$.

$\hat{y} =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q2/5

2067

QUESTION ID

S67-B2-Q2

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B2-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What does the least-squares line minimize?

A. The y-intercept

B. The sum of the squared residuals

C. The number of points

D. The slope

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q3/5

2068

QUESTION ID

S67-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The sum of the squared residuals

GRADING NOTE That is what "least squares" means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q3/5

True or false?

The least-squares line passes through the point $(\bar{x}, \bar{y})$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q4/5

2069

QUESTION ID

S67-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is a property of the least-squares line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B2-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Two people drawing lines of fit by eye always get the same line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q5/5

2070

QUESTION ID
S67-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Lines drawn by eye vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSS-ID.C.8

FORWARD 1

Use simulation to judge whether a correlation could be due to chance • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In 1,000 random shufflings of the y-values, r was at least 0.9 in 2 trials. The observed r is 0.92.

Is the association likely to be due to chance?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q1/5

2072

QUESTION ID
S67-F1-Q1

TYPE
Short answer

CORRECT ANSWER
No. A correlation this strong happened only about 0.2% of the time by chance.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For a different data set, shuffled data gave r ≥ 0.3 in 350 of 1,000 trials, and the observed r is 0.3.

What is the conclusion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q2/5

2073

QUESTION ID

S67-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The association could easily be due to chance (about 35%).

GRADING NOTE Any equivalent conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q2/5

Why shuffle the y-values in such a simulation?

A. To increase r

B. To see how large r tends to be when there is no real association

C. To remove outliers

D. To make the data linear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q3/5

2074

QUESTION ID
S67-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. To see how large r tends to be when there is no real association

GRADING NOTE Shuffling breaks any real connection between x and y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F1-Q3**
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q3/5

True or false?

With a larger sample, a moderate correlation is less likely to be due to chance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q4/5

2075

QUESTION ID
S67-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Chance correlations shrink as samples grow.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q4/5

True or false?

If shuffled data often give an r as large as the observed r, the association is significant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q5/5

2076

QUESTION ID
S67-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Then it could easily be chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSS-ID.C.8

FORWARD 2

Evaluate claims based on a correlation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A news story says, "r = 0.6, so 60% of the change in sales is caused by advertising."

Name two errors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

2078

GRADING NOTE Form required: two errors.

QUESTION ID
S67-F2-Q1

TYPE
Short answer

CORRECT ANSWER
r is not a percent of change ($r^2 \approx 0.36$ describes the share of variation explained), and correlation does not show causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A study of 8 cities finds r = 0.7 between the number of parks and happiness scores.

Why should you be cautious?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

2079

QUESTION ID
S67-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The sample is small, the study is observational, and other variables (such as income) could explain both.

GRADING NOTE Any two valid reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which statement about r = −0.9 is correct?

A. There is a strong negative linear association.

B. The slope of the line is −0.9.

C. 90% of the points are on the line.

D. An increase in x causes y to fall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

2080

QUESTION ID
S67-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. There is a strong negative linear association.

GRADING NOTE r measures the direction and strength of a linear association only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

True or false?

A correlation can be strong even when the relationship is not causal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

2081

QUESTION ID
S67-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A lurking variable can create it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

True or false?

If r = 0.8, the slope of the least-squares line is 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

2082

QUESTION ID
S67-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The slope depends on the units and spreads of x and y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

SET 68 • HSS-ID.C.9 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Distinguish between correlation and causation

MAIN	Algebra I main question	**HSS-ID.C.9**
BACKWARD 1	Interpret the correlation coefficient	**HSS-ID.C.8**
BACKWARD 2	Use random samples to make valid generalizations	**7.SP.A.1**
FORWARD 1	Distinguish surveys, experiments, and observational studies	**HSS-IC.B.3**
FORWARD 2	Use random methods to make fair decisions	**HSS-MD.B.6**

SET 68 • HSS-ID.C.9

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Cities with more firefighters have more fire damage.

Does hiring firefighters cause damage? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q1/5

2085

QUESTION ID

S68-M-Q1

TYPE

Short answer

CORRECT ANSWER

No. Larger cities (or larger fires) have both more firefighters and more damage; the city size is a lurking variable.

GRADING NOTE Any equivalent explanation naming a lurking variable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Ice cream sales and sunburn rates are strongly correlated.

Name a lurking variable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q2/5

2086

QUESTION ID
S68-M-Q2

TYPE
Short answer

CORRECT ANSWER
Hot, sunny weather

GRADING NOTE Any reasonable lurking variable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How could you test whether a new study app causes higher test scores?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q3/5

2087

QUESTION ID
S68-M-Q3

TYPE
Short answer

CORRECT ANSWER
Run a randomized experiment: randomly assign students to use the app or not, then compare the groups' scores.

GRADING NOTE Any design that uses random assignment and a comparison group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q3/5

Which study design can show that one variable causes changes in another?

A. A survey of volunteers

B. A scatter plot of existing data

C. An observational study

D. A randomized experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q4/5

2088

QUESTION ID

S68-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A randomized experiment

GRADING NOTE Random assignment balances other variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q4/5

True or false?

A strong correlation between two variables proves that one causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q5/5

2089

QUESTION ID

S68-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Correlation alone never proves causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q5/5

SET 68 • HSS-ID.C.9

BACKWARD 1

Interpret the correlation coefficient • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

ANSWER KEY Reduced copy of the question page, with its answer

r = 0.88.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q1/5

2091

QUESTION ID

S68-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A strong positive linear association

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which shows the stronger linear association, r = −0.7 or r = 0.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q2/5

2092

QUESTION ID

S68-B1-Q2

TYPE

Short answer

CORRECT ANSWER

r = −0.7

GRADING NOTE Any equivalent form. |−0.7| > |0.5|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B1-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is r for points that lie exactly on a line with negative slope?

A. 1

B. 0

C. −0.5

D. −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q3/5

2093

QUESTION ID

S68-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −1

GRADING NOTE A perfect negative linear relationship.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The correlation coefficient has no units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q4/5

2094

QUESTION ID

S68-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is computed from standardized values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q4/5

True or false?

Switching the roles of x and y changes the value of r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q5/5

2095

QUESTION ID

S68-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE r is the same either way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSS-ID.C.9

BACKWARD 2

Use random samples to make valid generalizations • Grade 7 • Statistics & Probability • 7.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Why is a random sample more likely to represent a population?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q1/5

2097

QUESTION ID

S68-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Every member has the same chance of being chosen, which avoids bias.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q1/5

A survey at a gym asks about the exercise habits of all town residents.

What is the problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q2/5

2098

QUESTION ID

S68-B2-Q2

TYPE

Short answer

CORRECT ANSWER

People at a gym probably exercise more than other residents, so the sample is biased.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a random sample of a school's students?

A. Drawing 40 names from a list of all students

B. Asking the first 40 students in line

C. Asking your friends

D. Asking one math class

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q3/5

2099

QUESTION ID

S68-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Drawing 40 names from a list of all students

GRADING NOTE Only the drawing gives every student the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q3/5

2099

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Generalizations about a population are valid only if the sample represents the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q4/5

2100

QUESTION ID

S68-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is why random sampling matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A sample of volunteers is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q5/5

2101

QUESTION ID

S68-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volunteers choose themselves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSS-ID.C.9

FORWARD 1

Distinguish surveys, experiments, and observational studies • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Researchers record students' screen time and grades without changing anything.

What kind of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q1/5

2103

QUESTION ID
S68-F1-Q1

TYPE
Short answer

CORRECT ANSWER
An observational study

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Researchers randomly assign plants to receive fertilizer or no fertilizer.

What kind of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q2/5

2104

QUESTION ID
S68-F1-Q2

TYPE
Short answer

CORRECT ANSWER
An experiment

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the main purpose of a survey?

A. To test whether a treatment causes an effect

B. To assign treatments to subjects

C. To control lurking variables

D. To estimate a characteristic of a population from a sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q3/5

2105

QUESTION ID
S68-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. To estimate a characteristic of a population from a sample

GRADING NOTE A survey collects information without assigning treatments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q3/5

True or false?

Random selection lets you generalize to a population; random assignment lets you conclude cause and effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q4/5

2106

QUESTION ID
S68-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The two kinds of randomization do different jobs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q4/5

True or false?

In an observational study, the researchers decide which treatment each subject gets.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q5/5

2107

QUESTION ID
S68-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They only observe.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSS-ID.C.9

FORWARD 2

Use random methods to make fair decisions • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Describe a fair way to split 20 volunteers into two groups of 10 using random numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q1/5

2109

QUESTION ID
S68-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Number the volunteers 1 to 20, randomly draw 10 different numbers for group A, and put the rest in group B.

GRADING NOTE Any method that gives every volunteer the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A coach picks a captain by drawing one name from a hat with 12 names.

What is the probability that a given player is chosen?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q2/5

2110

QUESTION ID

S68-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which method is fair for deciding who goes first among 4 players?

A. The tallest goes first

B. Each rolls a number cube; ties roll again; the highest goes first

C. The coach's favorite goes first

D. Whoever asks first goes first

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q3/5

2111

QUESTION ID
S68-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. Each rolls a number cube; ties roll again; the highest goes first

GRADING NOTE Only the cube roll gives everyone the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q3/5

True or false?

A fair lottery gives every participant the same probability of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q4/5

2112

QUESTION ID
S68-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE That is what fair means here.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Flipping one fair coin once is a fair way to choose one person from 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q5/5

2113

QUESTION ID
S68-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A coin has only 2 equally likely outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q5/5